

The Incidence of Carbon Taxes in an Electrifying Economy

Sevim Dinlemez

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Abstract

I show that when taking into account household electrification, the incidence of carbon taxes becomes age-specific and falls hardest on middle-aged, low-income households. In addition, it creates an equity–efficiency trade-off in the choice of revenue recycling instrument. In the paper, I develop a life cycle model with a discrete household electrification decision subject to financing frictions that slow adoption among low-income households, and with energy as a necessity good. The government levies a carbon tax to reduce emissions to an exogenous climate target, and recycles revenues via government spending, lump-sum rebates, renewable-electricity subsidies, or household electrification subsidies. Using renewable-electricity subsidies reduces the required peak carbon tax from roughly €340 to €210 per ton of CO₂ compared to all other instruments, halving the long-run GDP loss. However, the negative welfare effects concentrate among low-income, middle-aged households, where the cost of electrifying and the cost of not electrifying intersect. On the contrary, lump-sum transfers provide insurance against the burden from electrification for low-income households, albeit at the cost of larger GDP losses. Front-loading a carbon tax while recycling revenues with subsidies hurts poor households via the electrification margin, but not with lump-sum transfers.

Keywords: Carbon Tax, Welfare, Inequality, Electrification, Revenue recycling

1 Introduction

Recycling carbon-tax revenue into green subsidies makes the transition cheaper for the economy as a whole; a lower carbon tax suffices for a given climate target. Most of that gain, though, flows to households who are rich enough to electrify and reap the benefits of green subsidies, while the poor are better off receiving the revenues as a lump-sum transfer. For households, electrifying means replacing a gas boiler and a petrol car with a heat pump and an electric vehicle, and that switch requires an upfront payment that poorer households struggle to make (Berg et al., 2024). The efficiency-enhancing subsidies flow to rich households that would have electrified even without subsidy, while households that do not switch, either because switching is not optimal or because they cannot afford it, pay the tax in full. In this paper, I quantify how household electrification shapes the distributional incidence of a carbon tax, and how, together with electricity sector decarbonization, the recycling instrument affects the trade-off between efficiency gains at the cost of the unequal incidence.

This paper makes three contributions. First, I show that the incidence of a carbon tax is heterogeneous with age, and falls hardest on middle-aged, low-income households. Second, I show that while a green subsidy potentially halves the losses to GDP caused by a carbon tax, it is regressive: the electrification margin accounts for up to 60% of the welfare loss of low-income households. On the other hand, lump-sum recycling does not attenuate any GDP losses, but benefits low-income households. I thus characterize an equity–efficiency trade-off in the choice of revenue-recycling instrument that emerges only when household electrification and electricity-sector decarbonization are modeled jointly. Third, I show that the effect of front-loading the carbon tax on currently alive generations is ambiguous and depends on the revenue recycling instrument. While lump-sum recycling makes front-loading welfare enhancing, it decreases welfare further under green subsidy recycling. The latter is especially true for low-income households, caused by the electrification margin.

First, I can assess the heterogeneous age incidence of a carbon tax because I model electrification as a discrete, forward-looking investment in a life cycle model. I show that the carbon tax incidence depends on age, and middle-aged, poor households bear the largest burden. Since households recover the upfront cost of electrification only later, through lower energy bills, what matters for the electrification decision is not income and wealth alone but the combination of income, wealth, and age. The combination of a life cycle and a discrete, costly electrification margin creates a hump-shaped incidence of the tax over the age distribution for low-income households. Models without the life-cycle structure and extensive margin miss this hump-shaped pattern, and underestimate how concentrated the tax incidence actually is in different age groups. This matters for the politics of carbon taxes too: households between 40 and 65 are consistently among the strongest opponents of the policy (Levi, 2021).

The main mechanism in the model that creates this hump-shaped tax incidence comes from the fact that for middle-aged, low-income households switching to a green durable is as costly as paying the carbon tax for their remaining

lifetime. Households who do not electrify are directly exposed to the carbon tax, and as energy is a necessity, they cannot substitute away via the intensive margin. On the other hand, households who do electrify pay a large upfront cost. The large upfront payment is concentrated and near-term, while pay-offs only materialize via lower energy expenditures in the future. Thus while the costs of electrifying, i.e. user-cost of the durable good, increases in age, the costs of not electrifying, i.e. the remaining energy expenditures, decrease in age. The age group affected most by the carbon tax via the electrification margin, is at the age of intersection of these costs.

My second contribution is to characterize an equity–efficiency trade-off in the choice of revenue recycling instrument which only emerges when household electrification and electricity-sector decarbonization are modeled jointly. The supply side determines the *size* of the distortions caused by carbon tax losses and the potential subsidy gains; the electrification margin determines *who* receives those losses and gains. Green subsidies lower the peak carbon tax required for an exogenous target, and thus reduce aggregate GDP and wage distortions, but because richer households electrify earlier, the subsidies accrue mostly to them via lower electricity prices or higher electrification subsidies. Shifting revenue toward lump-sum redistribution lowers long-run GDP, but improves welfare for low-income households, who receive a larger share of the tax revenue directly. Models that assess the detailed effects of either margin alone, do not capture this trade-off (Ascari et al., 2025; Hochmuth et al., 2025; Kuhn & Schlattmann, 2024; Crucitti et al., 2024).

The model makes the trade-off quantitative. Households differ in income (à la Aiyagari (1994)), wealth, age and electrification status. Electrification is a discrete, forward-looking decision modeled as state variable: a household with a brown durable can pay a large upfront cost, potentially using a collateralized loan, to switch to a green durable. The type of durable determines the energy source (fossil fuels for the brown, electricity for the green) and converts it into services of mobility and heating. A representative final-goods firm combines capital, labor, and an energy composite of electricity and imported fossil fuels; the electricity sector aggregates renewable and fossil generation.

I examine three ways via which the government can ease the burden of a carbon tax using revenue recycling instruments, relative to wasteful government spending: (1) lump-sum transfers, (2) renewable electricity subsidies, and (3) electrification subsidies. For each instrument, I estimate the carbon tax that achieves a target emissions path and abstract from climate damages so that the instruments are compared internally on equal terms. The efficiency case for the subsidy is real, but narrow. Subsidizing renewable electricity lowers the required carbon tax and electricity price that long-run GDP falls only half as much as under lump-sum recycling. An electrification subsidy produces no comparable gain, since it barely affects energy prices. On equity, electricity and electrification subsidies impose the largest welfare losses on middle-aged, low-income households, almost entirely through the electrification margin following the above mechanism; lump-sum redistribution fully offsets this burden by offering broad consumption insurance to these households.

Third, I continue by showing that the effect of front-loading the carbon tax on the equity-efficiency trade-off varies per instrument. In all cases, front-loading makes emissions curb sooner, which implies that a lower peak tax rate suffices to reach the same climate target which reduces GDP decreases. The welfare burden it imposes differs quantitatively per instrument. A more rapid implementation of the carbon tax combined with green subsidies amplifies welfare losses under the existence of the electrification margin. When recycling revenues with green subsidies, the higher tax in the short run hurts poor households who are unable to adapt to the carbon tax early by electrifying. Pairing front-loading with lump-sum redistribution reverses this: transfers are higher in the short run, raising welfare for low-income households.

The remainder of this paper is structured as follows. I first relate this paper to the existing literature in section 1.1. In section 2, I discuss the model that I parametrize in section 3. In section 4, I conduct policy experiments and analyze the results for equity and efficiency. I analyze how the results change when I front-load the tax in section 5, and conclude in section 6.

1.1 Related literature

A growing quantitative literature studies inequality along the green transition. Earlier work treats energy purely as an input into production and traces inequality through labor income and revenue redistribution (Benmir & Roman (2022); Boehl et al. (2024)). Closer to this paper, other work treats energy as both a production input and something households consume directly. Känzig (2023) models energy as a fossil fuel that enters both production and a non-homothetic utility function, so that a carbon tax hits households twice: indirectly, through lower income, and directly, through their own energy use. Fried et al. (2024) use a similar non-homothetic setup to compare carbon-tax rebate instruments, and find that cutting distortionary taxes and raising the progressivity of labor taxes raises welfare more than lump-sum transfers do. The main difference with my paper is the differential types of energy, green and brown.

A second strand separates green and brown energy, so that climate policy can cut emissions by decarbonizing electricity itself rather than only by inducing households to consume less of it. Ascari et al. (2025) model two energy sources, fossil and renewable, that a profit-maximizing firm blends without friction into a single bundle sold to every household and firm; a carbon tax raises the fossil price and shifts that blend toward renewables. Because every household consumes the same bundle, inequality in their model comes entirely from non-homotheticity — energy’s share of spending falls as income rises. Hochmuth et al. (2025) build the same structure and show that the elasticity of substitution between green and brown energy, together with the degree of non-homotheticity, sets the size of the inequality effect. Both papers hold each household’s energy mix fixed across households; I let it vary by giving households an electrification choice, so a household’s exposure depends on whether it has switched, not only on its income. The composite electricity good in my model — a CES blend of renewable and fossil generation — follows the same structure as theirs.

Labrousse & Perdureau (2024) and Schlattmann (2024) add a spatial margin to the same green-brown distinction: households choose where to live, and rural households, who rely more on fossil energy, bear a larger burden from a carbon tax. Schlattmann (2024) also lets households choose between a green and a brown car. Where their margin is location, mine is the upfront cost of electrifying in place.

The electrification margin itself is the focus of Kuhn & Schlattmann (2024), who study, in partial equilibrium, how a subsidy for switching from an old, more polluting durable to a modern one affects inequality and transition speed under different financing schemes. Crucitti et al. (2024) study heat-pump adoption over the life cycle under a similar upfront cost, and find that a proportional subsidy benefits older households more than a lump-sum one does, because they own larger homes; Schlattmann (2024), discussed above, finds the same pattern for house size. I set aside these size effects and instead make the discrete decision to switch technology the central channel through which a carbon tax affects inequality. None of these papers lets the electricity sector itself decarbonize. Letting electricity supply respond too is what separates the size of the gains a subsidy creates from who actually receives them. Fried (2026) studies a related adoption margin driven by climate damages rather than climate policy: richer households adopt heat pumps to manage temperature extremes. The technology choice is the same; what triggers it differs — a carbon tax here, rising temperature variance there.

Finally, Fried et al. (2018) compare steady states to study how the burden of a carbon tax falls across generations under different recycling rules. I instead follow households through the transition itself, where the adoption friction creates a gap between the short and long run that a steady-state comparison cannot show.

2 Model

Overlapping generations of households populate the economy. During the work-stage the household faces idiosyncratic productivity shocks. On top of a standard consumption-savings model, I add two novelties that allow me to analyze the effect of an electrification margin. First, I add energy consumption by the household and include a subsistence requirement to capture the non-homothetic preference for energy services. As a novelty, I differentiate between two sources of energy, fossil fuels and electricity. The second addition is a durable good that governs the type of energy the household consumes. A household owns either a brown durable, consuming fossil fuels, or a green durable, consuming electricity. When switching from a brown to a green durable, hence electrifying, the household can use a collateralized loan. The durable good and loan are modeled as state variables. Throughout this section, all state variables chosen in the current period and carried into the next are denoted by a prime.

On the production side there are three sectors: a final-consumption good sector, an energy sector and, a durable goods sector. Lastly, a government

collects carbon tax revenues and recycles them within the same period.

2.1 Households

Demographics Time is discrete. Households' age is indexed by $j = 0, 1, 2, \dots, J$. After period $J - 1$ the household dies with certainty. A household works for J_w periods, and then retires. The economy holds a continuum of households with mass one. At birth, households differ in their income $y_{i,0}$, initial wealth $a_{i,0}$ and the durable good they are born with, $\phi_{i,0}$, which is a binary variable indicating whether the household owns a brown or green durable. Households are indexed by i , however for notational convenience this index is dropped in most cases.

Preferences Households maximize expected lifetime utility

$$U = E_0 \left[\sum_{j=0}^{J-1} \beta^j u(c_j, \mathcal{S}(\phi_j, e_j)) + \beta^J \mathcal{W}(b) \right], \quad (1)$$

where β is the discount factor, c is consumption of the final consumption good, $\mathcal{S}(\phi_j, e_j)$ is services derived from energy, and $\mathcal{W}(b)$ is a warm-glow bequest. Flow utility is a Stone-Geary utility function nested in a CRRA

$$u(c, \mathcal{S}(\phi, e)) = \frac{1}{1 - \sigma} (c^\gamma (\mathcal{S}(\phi, e) - \bar{\mathcal{S}})^{1-\gamma})^{1-\sigma}, \quad (2)$$

where γ denotes the households' relative taste for the final consumption good, $\frac{1}{\sigma}$ is the intertemporal elasticity of substitution, and $\bar{\mathcal{S}}$ is a subsistence level for services derived from energy. The subsistence level captures the non-homotheticity in energy demand.¹

The amount of energy services, $\mathcal{S}$, depends on the level of energy consumption, e , and the type of durable good, ϕ . A household owns a green durable, d_G , or a brown durable, d_B . For notational convenience I denote this by a binary variable, $\phi \in \{0, 1\}$, such that $d_\phi := \phi d_G + (1 - \phi) d_B$. The type of the durable good, ϕ , sets the rate at which energy translates into services, such that

$$\mathcal{S}(\phi, e) = \begin{cases} A_{s,G} e & : \text{if } \phi = 1 \implies d_\phi = d_G \\ A_{s,B} e & : \text{if } \phi = 0 \implies d_\phi = d_B. \end{cases} \quad (3)$$

Akin to the technologies in Fried (2026), each durable good transforms energy (here fossil fuels or electricity) into services of mobility and heating at a different

¹Most papers either use a subsistence level to capture this non-homotheticity Wöhrmüller (2024); Schlattmann (2024); Fried et al. (2024) or alternatively the flexible indirect utility function of Boppart (2014), adopted by among others Ascari et al. (2025); Hochmuth et al. (2025). Formally, the former is quasi-homothetic, as it gives linear marginal budget shares in the interior (away from the subsistence level) whereas the latter is fully non-homothetic. Quasi-homotheticity is sufficient to capture that poor households cannot substitute away from energy when the price increases.

rate. A higher rate implies a more energy efficient durable good. The energy efficiency of the durable good is thus pinned down by its productivity parameters $A_{s,G}$ and $A_{s,B}$ for the green and brown durable good respectively.²

A dying household receives utility from bequeathing its savings and durable good. this utility is

$$\mathcal{W}(b) = \eta \frac{(b + \underline{b})^{1-\sigma}}{1-\sigma}, \quad (4)$$

with bequest strength η , and where $\underline{b}$ denotes the extent to which bequeathing is a luxury good, as in De Nardi (2004). The size of the bequest is given by

$$b_t = a'_{J-1,t} + ((1 - \phi'_{J-1,t})P^{dB} + \phi'_{J-1,t}P^{dG}_t), \quad (5)$$

where $P^{d\phi}$ denotes the price of the durable good.

Durable good and energy The economy has two types of durable goods: a green durable d_G and a brown durable d_B . As explained above, each converts energy into services (mobility, heating) at a rate specific to that technology, $A_{s,G}$ or $A_{s,B}$.³ Furthermore, the green durable runs on electricity, while the brown durable requires fossil fuels. Switching from a brown to a green durable is therefore called electrification.

A newborn household receives either type of durable good via a warm-glow bequest. The probability of receiving the green durable depends on the household's initial income level and follows the dying cohorts' joint distribution of income and durables. A household electrifies by selling its brown durable for a scrap value χP^{dB} and buying a green durable for $P^{dG}_t(1 - s_{dG,t})$, where $s_{dG,t}$ is a subsidy for electrifying households. Electrification is an absorbing state: a household cannot switch back from green to brown. When the household chooses to electrify, it can use a secured borrowing scheme m , subject to a down-payment constraint

$$m' \leq \lambda^{LTV} P^{dG}_t (1 - s_{dG,t}). \quad (6)$$

The household pays an interest rate premium r_m on the loan and repays following an amortization schedule, so that the fraction of the loan due each period is

$$\pi(j, t) = \frac{(r_t + r_m)(1 + (r_t + r_m))^{J-j}}{(1 + (r_t + r_m))^{J-j} - 1}, \quad (7)$$

giving next periods loan $m' = (1 + r_t + r_m - \pi(j, t))m$. Each durable good requires maintenance $\delta_\phi = \phi\delta_G + (1 - \phi)\delta_B$, paid every period.

²These productivities are calibrated as the coefficient of performance, as defined by the International Energy Agency of the respective durable goods.

³This can be thought of as fossil fuels (or electricity) being translated into services of mobility and heating by an electric vehicle (heat pump) and gasoline vehicle (regular boiler).

The fuel price per unit of energy that the household faces depends on the durable good it owns. As noted, the green durable good relies on electricity, E with price P_t^E , whereas the brown durable good relies on fossil fuels, FF with price $P^{FF} + \tau_{c,t}$, where the government carbon tax $\tau_{c,t}$ raises the price of the latter.

Income and wealth Households supply their labor inelastically and receive income as an endowment each period. Labor income is denoted as $w_t Y_{j,i}$, where $y_{i,j}$ denotes labor productivity of a household i at age j , and is given by

$$\log(y_{j,i}) = \xi(j) + z_{j,i}. \quad (8)$$

such that labor productivity $y_{j,i}$ consists of a deterministic lifecycle component $\xi(j)$ and an individual component, $z_{j,i}$, representing productivity differences between households. During the first J_w years, the idiosyncratic part of labor productivity $z_{j,i}$ is stochastic and follows an AR(1) process,

$$z_{j,i} = \rho_y z_{j-1,i} + \epsilon_{j,i}. \quad (9)$$

After J_w periods, the household retires. It then deterministically receives a fixed proportion f_{ret} of its' last-period income as a pension for the remaining $J - J_w$ periods.

Households can save in a risk-free asset denoted by a , earning interest rate r_t every period. Unsecured borrowing is not allowed. Initial wealth is zero with probability p_0 . If initial wealth is non-zero, it is drawn from a univariate log-normal wealth distribution with variance $\sigma_{a_0}^2$, which integrates up to the savings bequeathed by dying households in that period.

2.1.1 Recursive formulation household problem

A household is born with initial wealth a_0 , income y_0 , and durable good ϕ_0 . Thereafter, it solves the following problem during its lifetime, so for $j = 0, 1, 2, \dots, J-1$, the household solves

$$V_j(a, \phi, m; y, P_t) = \max\{V_j^{NA}(a, \phi, m, y; P_t, \Omega_t), V_j^{Electrify}(a, \phi, m, y; P_t, \Omega_t) + \mu_{adj}\} \quad (10)$$

Where $P_t : \{r_t, w_t, P_t^E, P_t^{FF}, P_t^{dG}, P_t^{dB}, \tau_{c,t}\}$ denotes the price vector and $\Omega_t : \{\tau_{c,t}, \mathcal{T}_t, s_{ER,t}, s_{dG,t}\}$ the vector of government policies where $\mathcal{T}_t$ denotes a lump-sum transfer, and the carbon tax and subsidies are defined as above. I drop the time subscript for individual state and choice variables for readability. $\mu_{adj} < 0$ is a negative preference shock. It can be interpreted as a temporary dislike to change, distributed iid over time and households. It turns the adjustment decision into a probability rather than a discrete choice. The probability depends on the difference between $V_j^{Electrify}(\cdot) - V_j^{NA}(\cdot)$, but spreads the mass smoothly,

which helps both convergence and calibration. Details can be found in appendix B. The value function for a non-adjuster is,

$$V_j^{NA}(a, \phi, m, y; P_t, \Omega_t) = \max_{a', c, e} u(c, s) + \beta E[V_{j+1}(a', \phi, m', y'; P_{t+1}, \Omega_{t+1})], \quad (11)$$

subject to

$$w_t y + \mathcal{T}_t + (1 + r_t)a \geq c + a' + e[(1 - \phi)(P^{FF} + \tau_{c,t}) + \phi P_t^E] + \delta_\phi P_t^{d_\phi} + \pi(j, t)m, \quad (12)$$

$$a' \geq 0, \quad (13)$$

$$m' = (1 + (r_t + r_m) - \pi(j, t))m. \quad (14)$$

Equation 12 is the budget constraint: labor income, a lump-sum transfer and asset income must cover consumption, savings, energy expenditures, depreciation and any required secured loan repayments. The remaining equations give the unsecured borrowing constraint and the evolution of the loan repayment.

The value function for a household who chooses to electrify is

$$V_j^{Electrify}(a, \phi, m, y; P_t, \Omega_t) = \max_{a', m', c, e} u(c, s) + \beta E[V_{j+1}(a', \phi', m', y'; P_{t+1}, \Omega_{t+1})], \quad (15)$$

subject to

$$w_t y + \mathcal{T}_t + (1 + r_t)a + m' \geq c + a' + e\phi' P_t^E + \delta_\phi P_t^{d_\phi} + (P_t^{d_G}(1 - s_{d_G,t}) - \chi P^{d_B}), \quad (16)$$

$$a' \geq 0, \quad (17)$$

$$m' \leq \lambda^{LTV} P_t^{d_G}(1 - s_{d_G,t}), \quad \phi' = 1. \quad (18)$$

Again, equation 16 is the budget constraint: all income, existing of labor, a lump-sum tax and wealth income, plus the loan taken out should exceed at least consumption, savings, energy expenditure, depreciation costs and the purchasing cost of the green durable minus the resale value of the brown.

Lastly $V_J = \mathcal{W}(b)$ is as in equation 4.

2.2 Production

The production side of the economy has three sectors: a final consumption good sector producing Y^C , an electricity sector producing Y^E and, and a durable goods sector producing the green and brown durables Y^{d_G} and Y^{d_B} . The economy imports fossil fuels.

Final consumption good A representative firm produces a final good Y^C using capital K_C labor L_C , electricity E_C , and fossil fuels FF_C . The good

serves as either a consumption good or an input into durable good production. The production technology is a constant-returns-to-scale nested CES:

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1-\psi)^{\frac{1}{\nu}} (A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c})^{\frac{\nu-1}{\nu}} + \psi^{\frac{1}{\nu}} (M(E_C, FF_C))^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}}, \quad (19)$$

where $\psi \in (0, 1)$ is the share of the energy composite, $\nu > 0$ the elasticity of substitution between the capital-labor bundle and the energy composite, $\alpha_c \in (0, 1)$ the capital share, and A_{KC} capital-labor bundle productivity. The energy composite $M(E_C, FF_C)$ nests electricity and fossil fuels:

$$M(E_C, FF_C) = [\kappa^{\frac{1}{\iota}} (A_{EC} E_C)^{\frac{\iota-1}{\iota}} + (1-\kappa)^{\frac{1}{\iota}} (A_{FFC} FF_C)^{\frac{\iota-1}{\iota}}]^{\frac{\iota}{\iota-1}}. \quad (20)$$

where $\kappa \in (0, 1)$ is the electricity share in the energy composite, $\iota > 0$ the elasticity of substitution between electricity and fossil fuels, and A_{EC} and A_{FFC} the respective productivities. The firm takes the electricity price P_t^E , the carbon-tax-included fossil fuel price $(P^{FF} + \tau_{c,t})$, the wage w_t , and the rental cost of capital $(r_t + \delta_K)$ as given, and maximizes profits:

$$\Pi^C = Y^C - w_t L_C - (r_t + \delta_K) K_C - P_t^E E_C - (P^{FF} + \tau_{c,t}) FF_C. \quad (21)$$

In equilibrium profits are zero.

Electricity A representative firm produces electricity by aggregating renewable electricity Y^{E_R} and fossil-generated electricity Y^{E_F} . I use a similar technology to that in Ascari et al. (2025), however, given the capital intensity of the sector, I simplify by assuming labor is not a production input. Renewable electricity uses capital alone:

$$Y^{E_R}(K_{E_R}) = A_{K_{R,t}} K_{E_R}, \quad (22)$$

K_{E_R} is capital allocated to renewable generation and $A_{K_{R,t}}$ its productivity. The time subscript denotes that this productivity increases over time to capture an exogenous part of the energy transition. Fossil electricity combines capital K_{E_F} and fossil fuels FF_E via a CES technology:

$$Y^{E_F}(K_{E_F}, FF_E) = \left[\zeta^{\frac{1}{\omega}} (A_{KF} K_{E_F})^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} FF_E^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}}, \quad (23)$$

where $\zeta \in (0, 1)$ is the capital share, $\omega > 0$ the elasticity of substitution between capital and fossil fuels, and A_{KF} capital productivity in fossil generation. The firm aggregates these into total electricity output:

$$Y^E(K_{E_R}, K_{E_F}, FF_E) = \left[\alpha_R^{\frac{1}{\lambda}} (Y^{E_R})^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} (Y^{E_F})^{\frac{\lambda-1}{\lambda}} \right]^{\frac{\lambda}{\lambda-1}}, \quad (24)$$

where $\alpha_R \in (0, 1)$ and $\alpha_F = 1 - \alpha_R$ are the shares of renewable and fossil electricity, and $\lambda > 0$ the elasticity of substitution between them. The firm takes the electricity price P_t^E , the carbon-tax-included fossil fuel price $(P^{FF} + \tau_{c,t})$, and the rental cost of capital $(r_t + \delta_K)$ as given, and maximizes profits:

$$\Pi^E = P_t^E Y^E - (r_t + \delta)(1 - s_{E_R,t})K_{E_R} - (r_t + \delta_K)K_{E_F} - (P^{FF} + \tau_{c,t})FF_E, \quad (25)$$

where $s_{E_R,t}$ is a government subsidy on the capital cost of renewable generation. In equilibrium profits are zero.

Durable goods. Two representative firms produce the green and brown durable goods by converting the final consumption good (the numeraire, with price normalized to one) at conversion rates $A_{G,t}$ and A_B :

$$Y^{d_G}(C_G) = A_{G,t} C_G, \quad (26)$$

$$Y^{d_B}(C_B) = A_B C_B, \quad (27)$$

where C_G and C_B are the quantities of the consumption good used as inputs. The time subscript on $A_{G,t}$ denotes that green durable productivity increases over time, capturing cost reductions in the green technology. Each firm maximizes profits:

$$\Pi^{d_G} = P_t^{d_G} Y^{d_G} - C_G, \quad (28)$$

$$\Pi^{d_B} = P^{d_B} Y^{d_B} - C_B. \quad (29)$$

In equilibrium, profits are zero, which pins down durable good prices: $P_t^{d_G} = 1/A_{G,t}$ and $P^{d_B} = 1/A_B$.

2.3 Government

The government runs a balanced budget. It receives carbon tax revenues every period, which it uses for government expenditure, a lump-sum uniform transfer, a subsidy for capital costs to generate renewable energy generation or a subsidy for the costs of electrification, such that every period

$$\begin{aligned} G_t + \mathcal{T}_t + s_{E_R,t}K_{E_R,t}(r_t + \delta_K) + s_{d_G,t}P_t^{d_G} \sum_{j=0}^{J-1} \int (\phi' - \phi) d\mu_{j,t} \\ = \tau_{c,t}[FF_{HH,t} + FF_{E,t} + FF_{C,t}] \end{aligned} \quad (30)$$

2.4 Equilibrium

Given an initial distribution of types i over initial asset holdings $a_{i,0}$, productivity levels $z_{i,0}$, ages $j_{i,0}$, durable goods $\phi_{i,0}$, securitized loans $m_{i,0}$, the law of motion for idiosyncratic productivities as in equation 9 and, a set of policies $\Omega_t : \{\tau_{c,t}, \mathcal{T}_t, s_{E_R,t}, s_{d_G,t}\}$, the equilibrium is defined by a set of prices

$P_t : \{r_t, w_t, P_t^E, P^{FF}, P_t^{d_G}, P^{d_B}\}$ and a distribution μ_t over state variables $\mathbf{x} = (a, \phi, m, y, j)$ such that

1. Given prices, households solve the recursive problem as laid out in equations 10-18, where the is given by value functions $V_j^{NA}(a, \phi, m, y; P_t, \Omega_t)$, $V_j^{electrify}(a, \phi, m, y; P_t, \Omega_t)$ and policy functions $a'_j(a, \phi, m, y; P_t, \Omega_t)$, $c_j(a, \phi, m, y; P_t, \Omega_t)$, $e_j(a, \phi, m, y; P_t, \Omega_t)$, $\phi'_j(a, \phi, m, y; P_t, \Omega_t)$ and $m'_j(a, \phi, m, y; P_t, \Omega_t)$.
2. The final-output good firm maximizes profits as defined in equation 21, electricity firm maximizes profits as defined in equation 25, and the durable good firms maximize profits as defined in equations 28 and 29.
3. All financial wealth bequeathed and all durables bequeathed equals the initial, aggregate wealth of the new-born generation, that is, for wealth

$$\int a'(\mathbf{x})d\mu_{J-1,t} = \int ad\mu_{0,t}, \quad (31)$$

durable goods are passed on conditional on initial income level, such that the joint distribution of durables and income is similar in the dying and newborn generation.

4. The government budget constraint holds as defined in equation 30.
5. Prices are such that markets clear for

$$\text{Capital:} \quad \sum_{j=0}^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \quad (32)$$

$$\text{Labor:} \quad \sum_{j=0}^{J-1} \int y d\mu_{j,t} = L_C \quad (33)$$

$$\text{Gr. Durable:} \quad Y_t^{d_G} = \sum_{j=0}^{J-1} \int \phi \delta_G + (\phi' - \phi) d\mu_{j,t} \quad (34)$$

$$\text{Br. Durable:} \quad Y_t^{d_B} = \sum_{j=0}^{J-1} \int (1 - \phi) \delta_B - \chi(\phi' - \phi) d\mu_{j,t} \quad (35)$$

$$\text{Electricity:} \quad Y_t^E = \sum_{j=0}^{J-1} \int e \phi d\mu_{j,t} + E_{C,t} \quad (36)$$

and the aggregate resource constraint, derived in appendix E, clears by Walras law.

3 Parametrization

To assess the quantitative trade-offs behind climate policy during the green transition, I parametrize the economy so that its initial steady state matches

the pre-transition period. I focus on the European green transition, and calibrate to the year 2015, before the Paris agreement and when electrification was still very low.

Preferences Households are born at age 25, work until 65 and die at 75. This implies J is 50, and J_w is 40. Generations are equally sized. I set the intertemporal elasticity of substitution $1/\sigma$ to $1/2$, a standard value in the life cycle literature. I calibrate the remaining preferences parameters: discount factor, β , consumption share, γ , energy subsistence level, $\bar{S}$, bequest preference, η , and bequest luxury, $\bar{b}$, internally, using a simulated method of moments. The key moments I target to discipline these parameters are the mean savings in the economy, the average energy expenditure over total expenditure, the degree of non-homotheticity in energy expenditures, the change in saving holdings during the retirement phase and inequality in bequests. I discuss this in more detail in section 3.1.

Durable good and Energy The composite durable converts energy into heat and mobility services. I follow Kuhn & Schlattmann (2024) in defining two-thirds of the composite good as the car and the remaining part as the heating system, based on expenditure shares on energy for both components. I define the rate at which energy is translated into services — $A_{s,G}$ and $A_{s,B}$ for the green and brown durable good, respectively — as energy efficiency. I calibrate the efficiency levels of the two technologies separately for heat and for mobility, then aggregate them for the composite good.

Energy efficiency is the rate at which a unit of energy, fossil fuel or electricity, translates into useful energy by the technology at hand: how much useful energy a technology extracts from its primary (or, for electricity, secondary) energy source. Because of transformation losses or gains at the point of use, energy services consumed are not the same as final energy demand.

For heating, I use energy efficiency estimates from IEA (2022).⁴ They define useful energy as energy available to satisfy the needs of the user, and refer to this as energy services demand, similar to my model. In their report, they call the difference between useful energy and the energy used to run the technology as the coefficient of performance (COP), which directly translates into $A_{s,G}^H$ and $A_{s,B}^H$ in my model. They state that heat pumps have a COP of around 4, while the range is wide: industrial heat pumps have a COP around 3, and district heating networks with large heat pumps reach a COP of 8. I set $A_{s,G}^H$ to 4. Natural gas boilers, on the other hand, exhibit a COP of around 0.8, so I set $A_{s,B}^H$ to 0.8.

For mobility, I use estimates from the US Department of Energy from 2024.⁵ They find that for electrical vehicles, the energy conversion rate from energy

⁴The Future of Heatpumps report

⁵<https://www.energy.gov/eere/vehicles/articles/fotw-1360-sept-16-2024-typical-ev-87-91-efficient-compared-30-conventional>

Parameter	Description	Value	Internal	Target/source
<i>Households</i>				
<i>Preferences</i>				
β	Discount factor	0.932	yes	See table 2
$1/\sigma$	Intertemporal substitution	1/2	no	Standard value
γ	Taste for consumption	0.913	yes	See table 2
$\bar{S}$	Energy service subsistence level	6.45	yes	See table 2
η	Strength of bequest motive	32.76	yes	See table 2
b	Luxury of bequest	1.8	yes	See table 2
<i>Durable goods and energy</i>				
$A_{s,G}$	Energy efficiency d_G	1.82	no	See text
$A_{s,B}$	Energy efficiency d_B	0.47	no	See text
δ_G, δ_B	Depreciation	0.02, 0.01	no	Schlöter (2022)
χ	Resale value	0.27	no	Gilmore & Lave (2013)
$1 - \lambda_{LTV}$	Down payment requirement	0.1	no	Own assumption
r_m	Loan premium	0.03	no	Realistic risk premium
μ_{adj}	Location preference shock	-1.2	yes	See table 2
σ_{adj}	Scale preference shock	0.32	yes	See table 2
<i>Income and wealth</i>				
ρ_y	Persistence	0.98	no	DNB Household Survey
σ_ϵ	Std. dev. income	0.1	no	DNB Household Survey
σ_{a_0}	Std. dev. wealth at age 0	0.556	no	DNB Household Survey
p_0	Zero-wealth share at age 0	0.39	no	DNB Household Survey
<i>Production</i>				
<i>Electricity</i>				
α_R	Renewable share	0.5	no	Hochmuth et al. (2025)
λ	EoS green and brown electricity	1.8	no	Papageorgiou et al. (2017)
ζ	Capital share E_F	0.3	no	Own assumption
ω	EoS capital and fossil fuels in E_F	0.25	no	Coenen et al. (2024)
A_{KR}	Productivity green capital	14.52	yes	See table 2
A_{KF}	Productivity brown capital	16.26	yes	See table 2
<i>Durable goods</i>				
A_G	Productivity Y^{d_G}	0.86	no	Kuhn & Schlattmann (2024)
A_B	Productivity Y^{d_B}	2.2	no	Price of d_B
<i>Final consumption good</i>				
α_c	Capital share	0.33	no	Literature
ν	EoS capital-labor and energy	0.2	no	Realistically low, see text
ψ	Energy share	0.1	no	Hochmuth et al. (2025)
ι	EoS Electricity and Fossil Fuel	1.5	no	Papageorgiou et al. (2017)
κ	Electricity share	0.2	no	Papageorgiou et al. (2017)
A_{KC}	Productivity capital-labor	1.092	yes	Normalization w
A_{EC}	Productivity electricity	0.018	yes	See table 2
A_{FFC}	Productivity fossil fuels	0.00059	yes	See table 2

Table 1: Parameter values

input to useful energy, $A_{s,G}^M$, lies around 0.75.⁶ For a gasoline car, energy efficiency $A_{s,B}^M$ is lower, as it only translates a fraction 0.3 of total energy into useful energy and the remainder is lost as heat. This adds up to $(0.33 \times 4 + 0.67 \times 0.75) = 1.82$ for $A_{s,G}$, and $(0.33 \times 0.8 + 0.67 \times 0.3) = 0.47$ for $A_{s,B}$.

For other parameters, I follow the empirical literature. Following estimates from Schloter (2022), electric vehicles depreciate faster than gasoline vehicles. Estimates for heat pumps are diverging. I set the depreciation for the green durable good, δ_{d_G} to 2%, against 1% for the brown durable good, δ_{d_B} . For the resale value, χ , I follow a resale value for cars from Gilmore & Lave (2013) of 40% and 0% for heating systems; for the composite good, this implies a resale value of 27%.

I calculate the relative (to income) price of fossil fuels as a composite of the gasoline price and the natural gas price, for mobility and heating purposes respectively, converted to kWh.⁷ The price of one unit of fossil fuels then pins down the tax in percentage to price per tCO2 unit, as there is a one-to-one relation between emissions and units of fossil fuels used.⁸

For the parameters governing secured borrowing for the purchase of a durable good, I set the minimum down-payment requirement at 10%, a realistic value I assume directly. I also add a premium on loans to households, since such loans typically carry high interest rates.

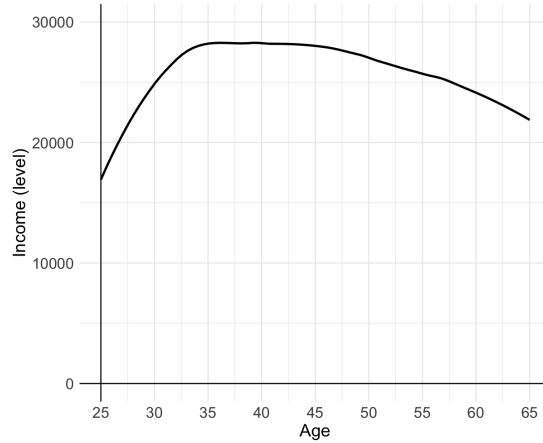


Figure 1: Deterministic life cycle component of productivity, $\xi(j)$

⁶taking into account cars with and without regenerative breaking

⁷For natural gas, Eurostat reports a price of 0.05 euros per kWh in Europe in 2015. (Online data code: nrg_pc_202) From the weekly oil bulletin, the price of gasoline (Euro-super95) is 529.7 euros per 1,000 liters. Using a conversion of 9.3 kWh per liter of gasoline, I calculate the price of gasoline at 0.0569 euros per kWh in Europe in 2015. Because the composite good is 67% mobility, the price of fossil fuels is 0.0546 euros per kWh. I define the model unit as 1,000 kWh and normalize it by mean income, which sets the price in the model to 0.0023.

⁸For example a tax of 150%. 150% of 0.0023 means 0.00345 per unit of FF, which implies $0.00345 \times 26033 = 89.81$ euros. One unit of FF emits 233 kCO2. Hence 4.3 units emit a ton of CO2. So the price is $89 \times 4.29 = 381$ euro per tCO2.

Income and initial wealth Income is determined by the endogenous wage w and exogenous, idiosyncratic productivity $y_{i,j}$. Productivity splits into a deterministic life cycle and a stochastic component. I normalize average productivity such that the average productivity equals one. This implies an average net labor income of 26033 euros in 2015.

I calibrate life cycle component ξ using the DNB household survey, by regressing the log of net income on age dummies and year fixed effects. Figure 1 shows the smoothed life cycle pattern of productivity. I then use the regression residual to calibrate the idiosyncratic, stochastic part $z_{i,j}$. Appendix A gives further details on the data used to calibrate of the income process.

For wealth data I again rely on the DNB household survey. Initial wealth is a combination of a degenerate distribution at zero wealth and a log-normal distribution over non-zero support, independent of income. I calibrate the fraction of households with a zero level of wealth, p_0 as those households with no or negative financial assets in the data at age 25. For the log-normal distribution, the mean is determined endogenously so that the value of assets held by the newborns integrates up to the value bequeathed by the dying households. I calculate the variance of the log normal distribution, $\sigma_{a_0}^2$, as the variance across all observations with positive financial assets at age 25.

Electricity firm Similar to Hochmuth et al. (2025), I assume α_R , the renewable electricity share in the aggregator, to be 0.5. I take estimates for the elasticity of substitution from the literature: 1.8 for the elasticity of substitution between green and brown sources in electricity generation λ , from Papageorgiou et al. (2017) and 0.25 for the elasticity between capital and fossil fuels in brown electricity production, ω , from Coenen et al. (2024). There is no consensus in the literature on the remaining parameter, ζ which governs the capital share in the production of E_F , and I set it to 0.3.

Lastly, I choose the productivity of capital in the electricity sector — for green capital A_{KR} and brown capital A_{KF} — to match (1) the share of renewable electricity and (2) the relative price of electricity over fossil fuels; I explain this in more detail in the internal calibration section below.

The results are robust to changes in ζ ranging from 0.1 to 0.5, as the parameter is relatively inconsequential due to the internal calibration of the productivity to match input shares.

Durable goods production I set the productivity level of the firm producing the brown durable good, A_B , such that the price of the brown durable good equals nearly half of the mean yearly income, namely 12000 euros. I set the price gap between the green and the brown durable to 1.9, similar to Kuhn & Schlattmann (2024), who calculate this for Germany.

During the transition, I let the price premium of the green durable shrink to zero by increasing the productivity $A_{G,t}$ in a linear fashion, such that at time

T , $P_T^{dG} = P_T^{dB}$. In order to determine T , I extrapolate the trend of the real price decrease in electrical vehicles and heat pumps over the period 2014-2020.

Final consumption good firm I follow the literature for both the capital share in the capital-labor composite α_c , and the the elasticity of substitution, ν , between capital-labor and energy. For the latter, the literature agrees on a realistically low elasticity of substitution ranging from 0.1 (Hochmuth et al., 2025; Fried et al., 2022) to 0.2 (Olovsson & Vestin, 2023). To stay conservative on the tax required to reduce industry emissions while remaining within the accepted range, I set ν to 0.2. Part of the results below depend on this parameter being below 1, a well-accepted value in the literature.

For the energy share, ψ , Hassler et al. (2021) show that this parameter is unimportant in this specification of the firm problem so long as ν is not close to one. For this parameter, Hochmuth et al. (2025) assume a random value of 0.5, and Ascari et al. (2025) discipline it internally. Because the parameter does not move much, and the productivities are calibrated internally conditional on it, I set this value to 10% given low estimates of energy expenditures out of total expenditures in the European economies. To pin down the elasticity of substitution between the different types of fossil fuels, ι , I use the empirical estimate from Papageorgiou et al. (2017), who estimate ι and the electricity share κ simultaneously at 1.5 and 0.2 respectively.

I internally calibrate the rates at which energy is transformed into the final good, productivities A_{EC} and A_{FFC} , to match the electricity input share and the share of total fossil fuels used in industry. Lastly, I set the productivity of capital-labor to normalize the wage to one in the initial steady state.

3.1 Internally calibrated parameters

To discipline 11 parameters that lack a clear data counterpart, I target several implied moments in the simulated data. For identification, I use 9 moments for the initial steady state; because μ_{adj} and σ_{adj} do not affect the initial steady state, I calibrate 2 moments from the early transition in a second step. Table 2 shows all moments. I match the model-implied moments with their data counterparts using a simulated method of moments. For the initial steady state, I choose a vector of parameters $\theta = (\gamma, \bar{S}, \beta, \eta, \underline{b}, A_{KR}, A_{KF}, A_{EC}, A_{FFC})$ in order to minimize the objective function

$$\min_{\theta} \sum_{i=1}^9 W_i \left(\frac{M_i^{model}(\theta) - M_i^{data}}{M_i^{data}} \right)^2, \quad (37)$$

in which W_i gives a weight for each moment i . There is not a one-to-one relation between the parameter and moment in table 2, as all parameters affect all moments. However, I denote in each row the parameter-moment combination that is most direct.

	Moment	Calculation	Data	Model
γ	Mean energy expenditure share	$\int_{i,j} P^{\phi_{i,j}} e_{i,j} / (P^{\phi_{i,j}} e_{i,j} + c_{i,j}) d\mu$	0.125	0.125
$\bar{S}$	Degree of non-homotheticity	$\int_{i,j Q1} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}} d\mu \bigg/ \int_{i,j Q5} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}} d\mu$	1.6	1.59
β	Mean wealth over mean income	$\int_{i,j} a_{i,j} d\mu \bigg/ \int_{i,j} w y_{i,j} d\mu$	3.6	4.42
η	Decay of wealth	$\int_i a_{i,J} d\mu \bigg/ \int_i a_{i,J_W} d\mu$	0.83	0.78
$\underline{b}$	Wealth inequality at death	$\int_{i D3} a_{i,J} d\mu \bigg/ \int_{i D6} a_{i,J} d\mu$	2.84	3.00
A_{KR}	Green electricity share	$\frac{E_R}{E_R + E_F}$	0.175	0.175
A_{KF}	Relative price of electricity	P_E / P_{FF}	2.40	2.40
A_{EC}	Input share electricity final good	$E_C / (E_C + F F_C)$	0.34	0.27
A_{FFC}	Final good fossil fuel share	$F F_C / (F F_C + F F_{HH} + F F_E)$	0.46	0.47
μ_{Adj}	Share electrified 2017	$\int \phi d\mu_{2017}$	0.012	0.012
σ_{Adj}	Price elasticity d_G 2017	$\frac{\partial \int \phi d\mu_{2017}}{\partial P^{d_G}} \frac{P^{d_G}}{\int \phi d\mu_{2017}}$	-1.36	-1.37

Table 2: Target moments in internal calibration

For the first and second moment, governing the energy expenditure share of households, I use the experimental statistics on income, consumption and wealth from Eurostat for 2015. This dataset uses data from the Household Finance and Consumption Survey. Using household-level data, it calculates expenditure shares for a variety of goods as well as total expenditures. In figure 2 the dotted gray line shows the energy expenditure share for households in each income quintile in the data. In the internal calibration, I match the average energy expenditure share of 12.56% and the degree of non-homotheticity as measured by the energy expenditure share in the first over the last quintile, equal to 1.6. I calculate the energy expenditure share over total expenditure, where the latter is defined as expenditures on all HCIP items. Matching the average and the degree of non-homotheticity pins down the energy expenditure per income quintile in the model results in the black line in figure 2.

For the moments related to the wealth and income on the household side, that discipline parameters $\beta, \eta, \underline{b}$, I use Dutch micro-data. The first three moments are calculated in the data using the DNB household survey, waves 2010 - 2020, I discuss this dataset in appendix A. For the discount factor β , I target the ratio of mean household financial net worth (NW) to mean net household income, the latter which the model normalizes to one. Second, for the strength of the bequest motive, η , I measure the mean decay of net worth in the data as the mean net worth at age 75 over the mean worth at age 65, in order to match

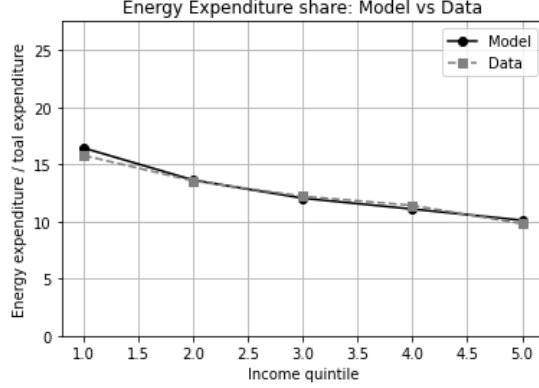


Figure 2: Energy expenditure share per income quintile in Eurostat data (2015) and in the model initial steady state.

the ages of retirement and death in the model. Lastly, to discipline the degree of luxury of bequests, $\underline{b}$ I target the degree of bequest inequality. I compute the mean financial net worth in the third and sixth decile at age 75, and take the ratio as the measure of inequality.

To discipline the productivities in the model, I use aggregate Eurostat data. I set the productivity of the green capital in electricity generation, A_{KR} to target the share of electricity that is produced from renewable sources, which in Europe in 2015 is 17.5%.⁹ To determine the productivity level of the brown capital in electricity generation, A_{KF} , I target the relative price of electricity over the price of fossil fuels. As explained above, the price of fossil fuels is 0.0546 per kWh. For the electricity price, I use the Eurostat DC-band price per kWh in Europe in 2015, which is 0.131 euros.¹⁰

I discipline the productivities for both energy sources in the production of the final consumption good, A_{EC} and A_{FFC} by matching the input share of electricity in the energy mix and the share of total fossil fuels used in industry. The Eurostat aggregate data for industry exclude the electricity sector but include all energy used by sectors that carry out other manufacturing and industrial processes. Measuring total fossil fuels used in industry as natural and manufactured gas, heat, oil and petroleum products, biofossils, and solid fossil fuels, I find that electricity makes up 34% of all energy used. To determine the amount of fossil fuels used in industry, I follow calculations from the Eurostat energy statistics report.¹¹ Of all fossil fuels used in the European economy, 46% went to industry (defined as manufacturing, storage, and services). The remainder consists of fossil fuels used in electricity production (21.8%) and by households for heating and mobility (26.2%). The remaining 6% is used for,

⁹Online data code: nrg_ind_ren

¹⁰Online data code: nrg_pc_204

¹¹Energy accounts, 2025

among other things, mining, water supply, and construction.

Lastly, in the second calibration step, the two remaining parameters, μ_{adj} and σ_{adj} are disciplined over the first years of the transition; they govern the aggregate rate of, and the heterogeneity within, households switching from a brown to a green durable. I discipline the mean and variance of the extreme value shock, μ_{adj} and σ_{adj} using a similar strategy used in Kuhn & Schlattmann (2024), namely by matching the share of households who switched to a green durable and the own-price elasticity of the green durable good. I target these moments after the transition has started. The share of households with an electrical vehicle in 2017 is 1.4% using aggregate statistics from statistics Netherlands (CBS). The percentage of households with a heat pump in that same year is 1.54%. For the elasticity, I rely on estimates of Fridstrøm & Østli (2021) who use Norwegian data to estimate the price elasticity of cars, among which electrical vehicles. In 2016, they find elasticities of 0.99 and 1.72 for battery electric and plug-in hybrid electric cars. Similar to the assumption in Kuhn & Schlattmann (2024), I take the average, which gives an elasticity of -1.36.

3.2 Baseline model validation

To validate the model, I evaluate the baseline switching behavior in the first years of the transition, and I compare inequality in the model and data. I focus on these two characteristics because they represent or relate to the main mechanism (electrification) and the main outcome variable (inequality).

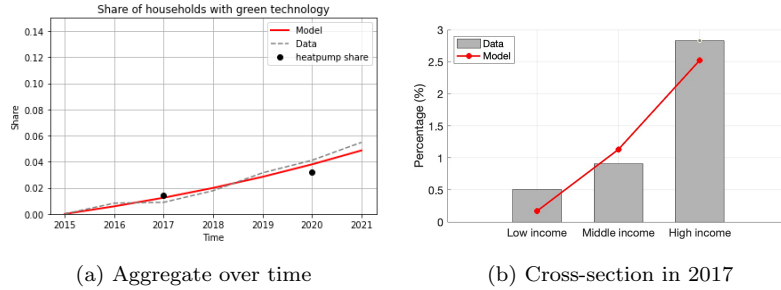


Figure 3: Electrification behavior over the transition

Switching behavior I use publicly available Dutch data from CBS on the share of homeowners with a heat pump and car owners with an electric vehicle. As in the model, the durable good is a composite of both, so I examine the realized time trends for each in the aggregate. Figure 3, panel (a), shows an overall good fit for uptake in the first years of the transition. 2015 is the year of introduction, so uptake starts at zero by default. The internal calibration targets the aggregate share of the composite good in 2017; the other years are untargeted, but the model still matches the trend, including in later years

Additionally, I analyze the heterogeneity in electrification across three income groups. I take a linear combination of the share of households with a

heatpump (33%) and the share of households with an electrical vehicle (67%) conditional on income: the lowest and highest quartiles, and the middle half. The percentages in the linear combination follow from the definition in the durable good. For heatpump shares, I use the same micro data as used for the figure in appendix A.2. For the uptake of electric vehicles, I use estimates from CBS, who estimate the share of EV owners in each income quartile.¹² They find that in 2021, from all EV owners, 5.5% is in the lowest income quartile, and 62.4% is in the highest income quartile. I multiply these numbers with the aggregate share in 2017, namely 1.2% to get to the share of EV-owners out of total households in each income group, and divide by the size of the respective income group. The results are shown in figure 3, panel (b). Overall, the untargeted distributional uptake of green durables fits the data well, both qualitatively and quantitatively. The model underestimates uptake in both the highest and lowest income groups, and slightly overestimates it in the middle-income group. The model's electrification pattern is slightly more linear in income than in the data, where the relationship appears somewhat more convex.

Inequality As I compare the distributional welfare effects among different income groups, it is necessary to properly match income inequality. In the EU the disposable income gini ranges from 23.7 to 37.9.¹³ I obtain a Gini coefficient of 36.8, near the upper end of this range.

4 Results

I solve the model using the algorithms laid out in appendices B - D. The economy starts in the initial steady state and unexpectedly enters a transition. In the baseline (no tax) transition, the green durable good becomes available at a price premium. The productivities of production of the green durable good A_G and green electricity A_{KR} increase over the course of the transition.¹⁴ I compare this baseline transition case to several policy experiments in which I introduce a carbon tax at the start of the transition, and will be phased in linearly. At the start of the transition, households learn the entire path of productivities, the carbon tax and recycling instrument and prices.

In the policy experiments, I increase the carbon tax τ_c from zero in time $t = 0$ to $\bar{\tau}_c^i$ in 25 years, and keep it fixed afterwards. I then compare efficiency (GDP) and distributional welfare outcomes against the baseline (no-tax) case by varying the instrument, i , used to recycle the tax revenues. I consider four revenue-recycling instruments: (i) government expenditure, G , (ii) a uniform lump-sum transfer, $\mathcal{T}$, (iii) a subsidy for green electricity production, s_{E_R} , and (iv) a subsidy for electrification of households, s_{d_G} , which I cap at 30%.¹⁵

¹²CBS report: who owns electric vehicles

¹³Eurostat, 2015

¹⁴I extrapolate the trends in price developments over the past 10 years and assign the same growth factor for the first 15 years of the transition. For the green durable productivity I make sure it converges to the brown durable's productivity A_B .

¹⁵I do this as the revenue including the firms' tax payments exceeds the amount that can

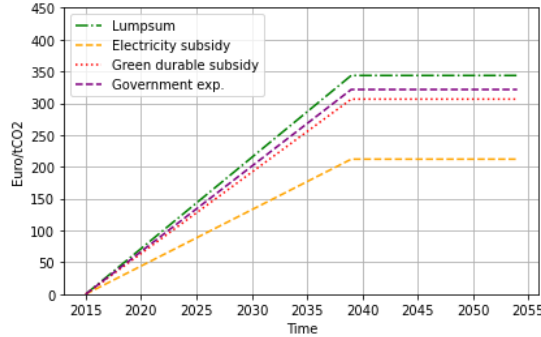


Figure 4: Carbon tax path by revenue-recycling instrument to meet the exogenous temperature target (2 degrees Celcius)

I compute the tax rate $\bar{\tau}_c^i$ that achieves a maximum temperature increase of 2 degrees Celcius above the pre-industrial level in 2065.¹⁶ I compute the development of temperature as a function of total cumulative emissions, using an off-the-shelf model discussed in appendix F.1, calibrated in F.2, which is exactly taken from Douenne et al. (2025). The figure for temperature development can be found in appendix F.3. An endogenous tax rate that reaches a given temperature allows for the efficiency benefits of green subsidies to materialize without explicitly modeling economic damages from temperature increases.¹⁷

Before examining the efficiency and distributional effects, I first highlight in figure 4 the endogenous tax paths for each instrument needed to achieve the temperature target. Using subsidies as a recycling instrument, especially a green electricity subsidy, reduces the peak tax rate required to achieve the temperature target. The reason for the lower tax rate with a green electricity subsidy lies in two mechanisms. First, with a subsidy for green electricity, for a given tax rate, the electricity sector substitutes more towards renewable generated electricity, so electricity production is less emission-intensive. Second, the subsidy for renewable electricity production lowers the electricity price compared with other instruments, so for a given carbon tax, the final-consumption-good producer substitutes more toward electricity than fossil fuels within the energy composite, M . The green durable subsidy has a smaller tax-reducing effect because households at the margin of switching in the baseline case are mostly poor and use little energy overall.

be given out as subsidies (that is tax revenues exceed all switching costs in early periods even if everyone where to switch). The 30% is in line with most taxes given out in Europe. In the default case, the remainder is spent on government expenditure.

¹⁶As the economy is not in net-zero at this point, the underlying assumption is that some carbon capture technology keeps the temperature at 2 degrees Celcius.

¹⁷Alternatively, I could set an exogenous tax rate and compute climate damages. In this case, climate damages would be lower with a subsidy rather than lump-sum redistribution or government expenditure due to lower emissions, which reduces GDP declines. In the current case I account for this by allowing for a lower tax rate under subsidy redistribution, also leading to lower GDP declines.

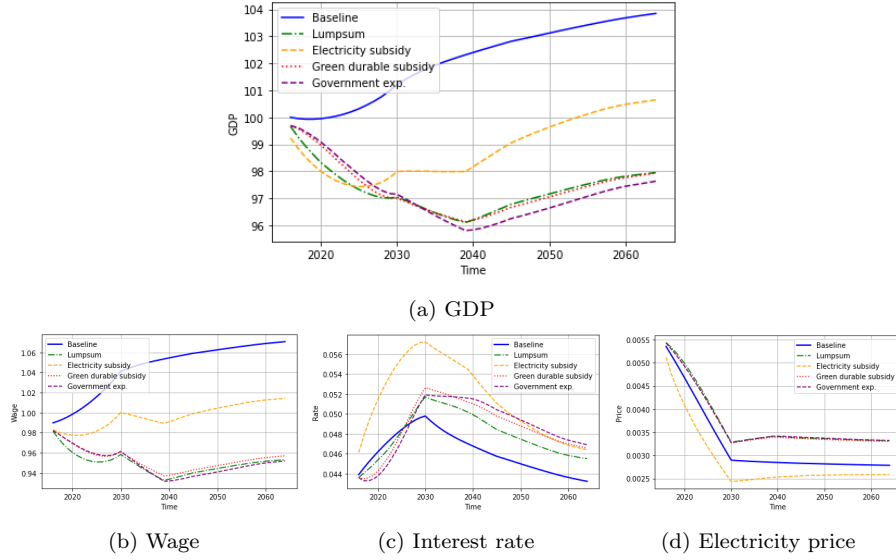


Figure 5: GDP (top) and prices (bottom) over the transition

4.1 Efficiency effects

To assess the extent to which the carbon tax distorts the economy, I assess the development of GDP and prices over the transition. Figure 5, panel (a), shows the development of GDP over the transition.¹⁸ GDP initially declines after the climate policy is introduced, for all instruments, because higher fossil fuel prices raise firms' production costs. However, the recovery of GDP after the initial dip depends heavily on the instrument used. Of the four instruments, the electricity subsidy mitigates GDP losses from the carbon tax the most, since it dampens distortions arising from higher energy prices.

In equilibrium, climate policy increases the interest rate (figure 5, panel (b)) and decreases the wage (figure 5, panel (c)). A shift from fossil energy imports (and thus final-consumption-good exports) toward electricity production — which is more capital-intensive than producing the final consumption good — raises capital demand and pushes up the interest rate. This effect is amplified in the short run, as households spend financial wealth on green-technology adoption that they would otherwise have saved.

Contrary to the interest rate, the wage decreases. Because labor is used only in producing the final consumption good, the wage is determined entirely within that sector. The price of the optimized energy composite, M , used in producing the final consumption good, increases for all recycling instruments. Because the capital-labor composite is complementary to the energy composite (elasticity of substitution $\nu < 1$), the wage drops. The muted wage drop under

¹⁸Discontinuities in the graph are explained by the phasing in of increased renewable electricity production for 15 periods (2030), and the tax for 25 periods (2040).

the electricity subsidy comes from the muted increase in the price of the energy composite M . The price of M increases less under an electricity subsidy for two reasons: first, the cost of fossil fuels is lower, following the lower endogenous tax rate estimated above; second, the electricity price itself is lower, as figure 5, panel (d) show. The same reasoning holds for the muted decrease in GDP in the electricity subsidy case.

4.2 Equity effects and the electrification margin

In figure 6 I show the consumption equivalence for the different recycling instruments, for the average household in a given age and productivity group (j, z) , for all households alive when the policy is introduced. With the exception of uniform lump-sum transfers as the recycling instrument, all agents are worse off from the carbon tax. Under lump-sum transfers, low-income households gain because the transfer raises their net disposable income, even after accounting for lower labor income, by more than their expenditures rise.

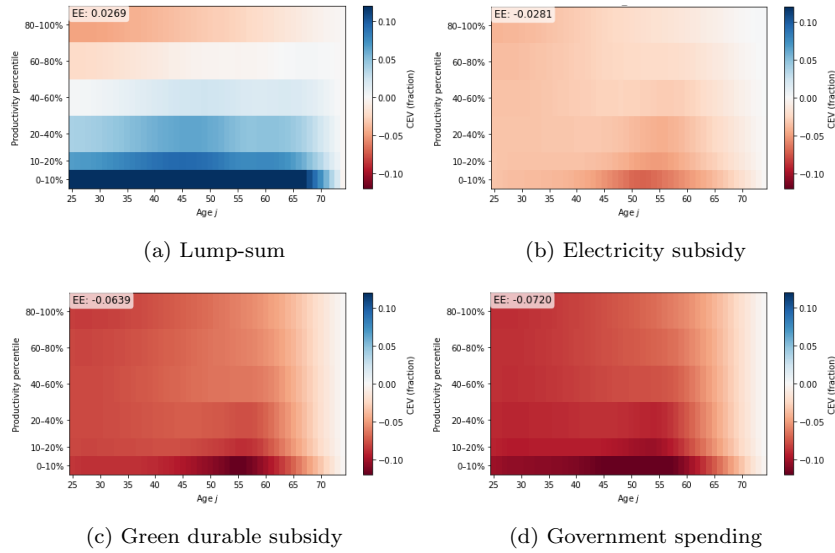


Figure 6: Consumption equivalence across the age–income space for different recycling instruments. Values indicate the percentage change in expenditure in the baseline required for households to be indifferent between the baseline and policy scenario.

In all three other cases all agents are worse off with carbon tax, and poor, middle-aged households suffer the largest welfare loss. Subsidies increase welfare compared with using tax revenues for government expenditure. When the subsidy targets renewable electricity, the welfare losses are smaller overall than under a green durable subsidy. This welfare difference comes from weaker negative general equilibrium effects under the electricity subsidy: as explained above,

the carbon tax and the price of electricity are significantly lower following the electricity subsidy, so wages decrease significantly less for the same carbon tax. Together, these two effects make electricity subsidies superior to a green durable subsidy in terms of welfare.¹⁹ However, the lower wage decrease and the drop in electricity prices mostly benefit rich, high-productivity households who electrify early. Put differently, the welfare gains of a green electricity subsidy concentrate mostly among rich households.

The role of electrification The electrification margin is the key driver of the high welfare loss for the low-income households. To quantify the role of the electrification margin in shaping the distributional welfare effects of the carbon tax, I compare the baseline consumption equivalent to one from a counterfactual economy in which all households are endowed with a green durable. Removing the extensive electrification margin in this way allows me to isolate the welfare consequences it causes after a policy introduction. I define the state-dependent welfare effect of the electrification margin as

$$\Delta CE^{Elec}(j, z) = CE_{t=1}(j, z) - CE_{t=1}^{noElec}(j, z),$$

where in the *noElec* case all households are endowed with green durables. In doing so, for each income, age group I subtract the welfare effect that would have arisen without electrification margin from the welfare effects plotted in figure 6. I plot this state-dependent welfare effect in Figure 7. A positive value implies that the electrification margin improves the welfare impact of the policy, while a negative value indicates that it causes a welfare loss following the policy introduction.

In all cases except lump-sum redistribution, the electrification margin makes households significantly worse off from a carbon tax. In contrast, under lump-sum redistribution, the electrification friction actually makes poor households slightly better off. This happens because, with the electrification margin, tax revenues — and therefore the lump-sum transfer — are higher at the start of the transition. However, in the three other cases, the electrification margin makes all households worse off following the policy introduction. The losses caused by the electrification margin are concentrated among the poor, middle-aged households. This shows that the electrification margin creates a heterogeneous impact of the carbon tax over the age distribution for lower-income groups. The hump-shaped pattern in the carbon tax incidence over the age distribution can be explained by an increase in expenditures on either energy, or durables. To show this, I will explain the mechanisms in terms of a change in expected lifetime consumption following the policy introduction.

¹⁹The combination of a green durable subsidy and electricity subsidy does not improve welfare much beyond what the electricity subsidy compared to the green durable subsidy does.

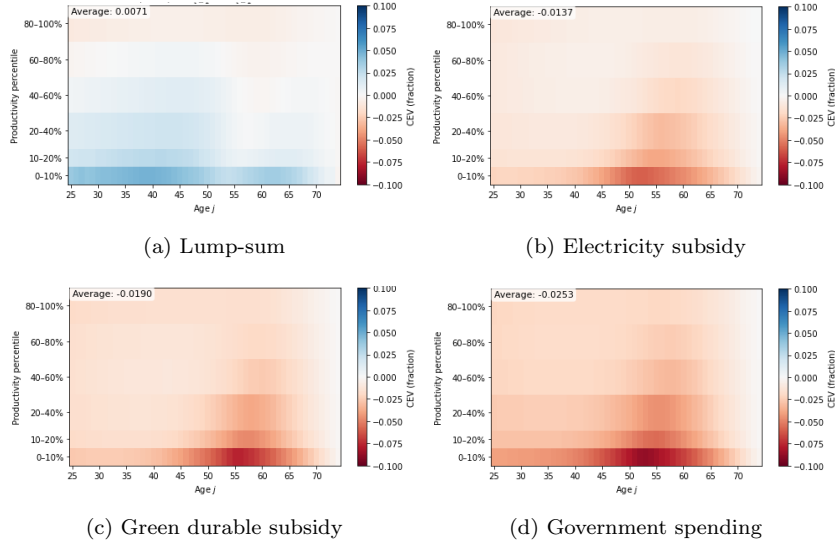


Figure 7: State-space heatmap of $\Delta CE^{Elec}(j, z)$ for different recycling instruments

4.2.1 Mechanisms

To understand the mechanisms underlying the distributional results, I compute and decompose the change in expected lifetime consumption, denoted by ΔLC , following the introduction of the policy. The welfare impact of this consumption change depends on sign and size of the change, the timing of the change (as the future is discounted), the degree to which the change is spread out over time, and the initial level of consumption. The concavity of the value function causes the latter two effects. For these reasons, the magnitude of the change in consumption translates nonlinearly into the magnitude of welfare effects. However, the sign of the consumption change has a one-to-one relationship with the sign of the consumption equivalent, and within income groups, the size remains indicative. Lastly, the biggest advantage of analyzing ΔLC is that I can use a forward-iterated lifetime budget constraint to decompose changes in ΔLC into five different components, which I formally define in proposition 1.

Proposition 1. *Consider household i , who is age $j_{0,i}$ at the time of the policy introduction. Define the remaining lifetime horizon as $S_i \equiv J - 1 - j_{0,i}$. For $s = 0, 1, \dots, S_i$, define household i 's lifetime consumption object as*

$$LC_i \equiv E \sum_{s=0}^{S_i} c_{i,j_{0,i}+s,s} + a_{i,J},$$

and define the five components

$$\begin{aligned}\text{Transfers}_i &\equiv \sum_{s=0}^{S_i} \mathcal{T}_s, \\ \text{LaborInc}_i &\equiv E \sum_{s=0}^{S_i} w_s y_{i,j_0,i+s}, \quad \text{AssetInc}_i \equiv E \sum_{s=1}^{S_i} r_s a_{i,j_0,i+s,s}, \\ \text{EnergyExp}_i &\equiv E \sum_{s=0}^{S_i} EE_{i,j_0,i+s,s}, \quad \text{DurableExp}_i \equiv E \sum_{s=0}^{S_i} DE_{i,j_0,i+s,s},\end{aligned}$$

where

$$\begin{aligned}EE_{i,j,t} &\equiv P_{\phi_{i,j,t},t} e_{i,j,t}, \\ DE_{i,j,t} &\equiv \delta_{\phi_{i,j,t},t} + f(j) m_{i,j,t} - (\phi'_{i,j,t} - \phi_{i,j,t}) (m'_{i,j,t} - P_t^G + \phi P^B).\end{aligned}$$

Then the change in household i 's expected lifetime consumption relative to the baseline admits the decomposition

$$\begin{aligned}\Delta LC_i &= \Delta \text{Transfers}_i + \Delta \text{LaborInc}_i + \Delta \text{AssetInc}_i \\ &\quad - \Delta \text{EnergyExp}_i - \Delta \text{DurableExp}_i.\end{aligned}\tag{38}$$

Proof. See appendix G. $\square$

I use this decomposition to answer two questions. First, how does the electrification margin shape the distributional incidence of climate policy? Second, why does lump-sum redistribution benefit low-income households?

The electrification margin and age-specific expenditures To understand why the electrification margin affects the middle-aged, poor households the most, I analyze the age heterogeneity in durable and energy expenditure, as defined in proposition 1, as this heterogeneity explains the red dark triangle in figure 7. In figure 8, I plot the changes in durable expenditures, energy expenditures and their sum for the lowest income group by paired age cohort, for lump-sum and electricity-subsidy recycling.

It is useful to separate three groups of households: (1) the young (roughly ages 25–35) households, who would have electrified in any case; (2) the old (roughly aged > 53 years old) households, who are not electrifying in any case; and (3) the switchers, the middle-aged (roughly ages 35–53) households, who would not have electrified without the carbon tax in place, but will do so with it. In group 1, the young households incur a small increase in durable expenditures as they electrify earlier with the policy in place, when P_t^{dG} is still higher. Under lump-sum redistribution, as the electricity price rises somewhat but proportionally less than fossil fuels, so do their energy expenditures. As a result, the effect on lifetime consumption through increased expenditures is limited for this group. In group 2, because these households are too close to the end of life to recover

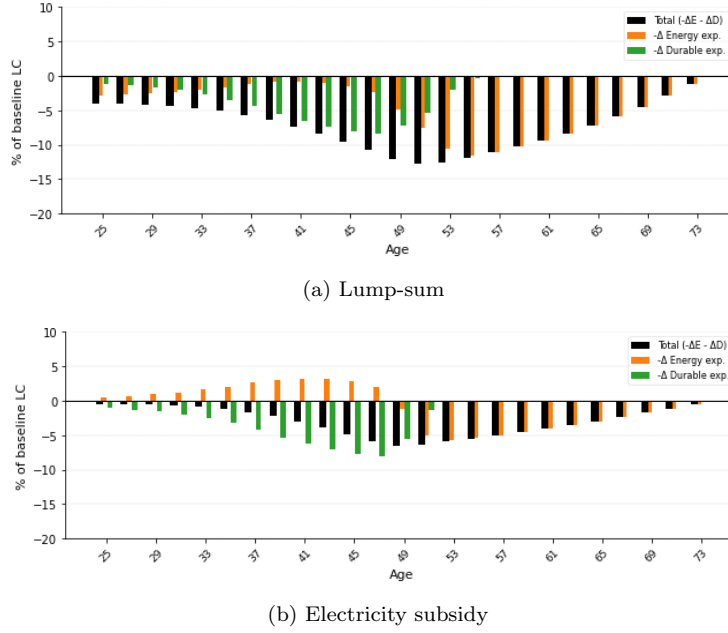


Figure 8: Heterogeneous expenditure responses across the age space for $z = 0$.

the fixed electrification costs, they optimally stick with the brown durable and consequently face a persistent increase in energy expenditures as fossil fuel prices rise. In group 3, the policy-induced switchers, electrification requires substantial upfront costs, generating a pronounced increase in durable expenditures. Whether these households are compensated through lower energy expenditures depends on the recycling instrument. Under lump-sum redistribution, electricity prices rise, so switching households face both higher durable expenditures and slightly higher energy expenditures. Under an electricity subsidy, electricity prices fall, which partly offsets the switching costs through lower future energy expenditures. Overall, changes in energy and durable expenditures hit the lifetime consumption of middle-aged households hardest (with the peak around ages 49–57, depending on the instrument), the ones close to the point of indifference between electrifying or not, once the carbon tax is in place.²⁰

The benefits of lump-sum transfers To understand why lump-sum is beneficial for poor households despite the higher tax rate on fossil fuels compared to the electricity subsidy, I plot the expected lifetime consumption change, ΔLC_i ,

²⁰Similar figures for the green durable subsidy and government expenditure can be found in Appendix H.1. The figure for government spending explains the triangle with the same mechanisms but increased magnitudes. The figure for green durable subsidy diminishes durable spending, which explains why the triangle in figure 7 panel (c) shifts to the right and thus the margin is especially salient for never-switchers.

as well as the five components, as a percentage share of total expected lifetime consumption absent any policy, for three income groups (conditional on age) in figure 9. The denominator is thus the same in both panels.

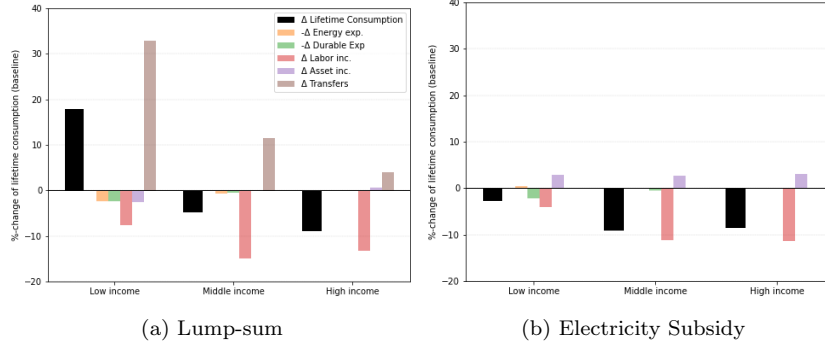


Figure 9: Decomposition of the change in expected lifetime consumption following policy introduction. Low-income refers to the lowest decile (conditional on age), high-income refers to the highest decile.

The largest part of the decrease in lifetime consumption comes from the decrease in wages. Comparing within-income-group variation between both instruments shows that, as expected, electricity subsidies soften the wage decline and the resulted loss in labor income. However, comparing across income groups within a given instrument, the same proportional wage decline costs high earners more in absolute terms, because a fixed percentage drop is larger in levels for higher incomes.²¹ Therefore, for low-income households, the lump-sum transfer is larger in magnitude than the combined effect of all other (negative) components, whereas for other income groups the overall change in lifetime consumption remains negative. This sign difference between income groups disappears under the electricity subsidy case, where low-income households have a small negative change in expected lifetime consumption. While low-income households are worse off with electricity subsidy compared to lump-sum redistribution, high-income households are slightly better off with electricity subsidies.

A carbon tax thus disproportionately raises durable and energy expenditures for middle-aged to old, low-income households, whether they switch technologies or remain on fossil fuels. Under lump-sum recycling, these expenditure burdens are larger, since both fossil fuel and electricity prices rise. Under the electricity subsidy, switchers benefit from lower electricity prices, and non-switchers face a smaller fossil-fuel price increase, so expenditure burdens are lower. Nevertheless, the financial relief for poor households is larger under lump-sum redistribution, given the size of the transfer relative to their lifetime consumption. In short, the

²¹This is inherit to a standard income process, where wage decreases are uniformly distributed. Others have assumed differently, however, see for example Känzig (2023) or Labrousse & Perdureau (2024). If it were assumed that low productivity households are more affected by wage drops or labor market loosening, this would negatively affect the case for lump-sum compared to electricity subsidies.

subsidy better targets the expenditure increases from the electrification margin as well as income decreases, but this mainly benefits rich households, while lump-sum transfers provide greater net income support to poor households.

The case without electrification In contrast to the findings of Ascari et al. (2025), who show that a carbon tax combined with an electricity subsidy is regressive, I do not find regressivity with an electricity subsidy but without the electrification margin. In their framework, regressivity arises from non-homothetic energy demand and the fact that the price of energy increases even when electricity is subsidized. The key difference is that, unlike in their setting, the price of electricity in my economy decreases when the policy is implemented with an electricity subsidy. As a result, even with non-homothetic energy demand, climate policy with revenue recycling through electricity subsidies is progressive. The reason for the declining electricity price is the assumption of continuous fuel switching of the final consumption good firm, which thus also uses fossil fuel. Because this firm pays the carbon tax on its fossil fuel usage, tax revenues increase. These additional revenues finance the renewable electricity subsidy, and the resulting subsidy-induced reduction in electricity prices more than offsets the upward pressure from the carbon tax on fossil fuels. Consequently, electricity prices fall, and policy effects turn progressive.

4.2.2 Combining lump-sum and electricity subsidies

From the above, we can establish that while green electricity subsidy mitigates losses in GDP by lowering energy prices, it is regressive via the electrification margin. On the other hand, lump-sum makes the carbon tax progressive at the cost of a higher GDP loss. In this section, I analyze whether combining electricity subsidies with lump-sum transfers can mitigate the trade-off between efficiency and equity. To assess this, I consider linear combinations of the two instruments, varying the share of tax revenues allocated to each. Table 3 reveals the central trade-off in this paper: subsidizing green electricity will relieve GDP distortions caused by higher energy prices, but as the resulting benefits mainly accrue to richer households, aggregate welfare losses and inequality increase. Both the efficiency and aggregate welfare results vary approximately linearly in the share of revenues allocated to lump-sum transfers versus electricity subsidies. Increasing the weight on lump-sum redistribution monotonically raises GDP losses while improving welfare (especially for low-income households), whereas shifting revenues toward electricity subsidies reduces GDP losses but worsens distributional outcomes.

The linearity of the outcome variables in the revenue shares implies that the two instruments operate through largely separable channels. Electricity subsidies primarily affect the economy through general equilibrium effects — lower electricity prices which reduce production distortions — while lump-sum transfers operate through direct redistribution. Because these channels interact only weakly, combining the instruments does not generate additional complementarities beyond a convex combination of their individual effects. As a result, policy

% Tax Rev. Subsidy (LS)	Future GDP Decrease (%)	Current EE Aggregate (%)	Current EE low-inc. (%)
100 (0)	-3.37	-2.81	-4.41
75 (25)	-3.74	-1.81	-0.11
50 (50)	-4.21	-0.60	3.11
25 (75)	-4.85	0.84	8.10
0 (100)	-5.80	2.69	14.83

Table 3: Policy trade-off between subsidy financing and lump-sum transfers

mixes trace out a clear efficiency–equity trade-off. Moving toward electricity subsidies improves aggregate efficiency but disproportionately benefits higher-income households who electrify early and therefore gain from lower electricity prices. In contrast, increasing the lump-sum component redistributes resources toward poor households, more than offsetting their higher energy expenditures, but at the cost of larger distortions from a higher carbon tax. These results imply there is no obvious interior optimum in this setting. The division of revenues remains a normative question that depends on the decision for the welfare function as well as the discount factor used for future generations.

4.3 Robustness and validity

A concern about the validity of the results is that I do not model economic damages from temperature increases, which complicates the comparison between the baseline and the policy scenarios. This does not affect the comparison across policy instruments themselves, since I estimate the carbon tax path so that temperature changes do not diverge across them. Therefore, the conclusions I draw apply only to comparisons across policy instruments, not to comparisons between any policy scenario and the baseline.

Moreover, four assumptions matter for the results and are natural candidates for robustness checks. First, idiosyncratic risk burdens households, but lump-sum redistribution partly offsets it, since deterministic transfers smooth consumption; removing idiosyncratic risk, while holding the income grid fixed so that income differences reflect only initial heterogeneity, does not change the distributional welfare results much. Second, the elasticity of substitution between energy and the capital-labor composite in the final consumption good sector, ν , sets the size of the wage decrease the tax causes; the resulting distortions are robust to increases in this elasticity when I hold tax rates constant. Third, the elasticity of substitution between renewable- and fossil-generated electricity, λ , largely governs the tax rate needed to limit the temperature increase under each instrument; the tax rate is sensitive to this elasticity, but the instruments’ relative ranking is not. Fourth, the minimum down payment required to purchase the green durable, λ^{LTV} , governs the severity of the financing friction; the results are robust to loosening the borrowing constraint, since households who do not electrify early are held back not by the constraint itself but by the

fact that adopting early is not profitable for them. Part of this is caused by rising opportunity cost via the increasing interest rate. The figures supporting these results as well as a more extensive description of the results can be found in appendix I.

5 Front-loading the carbon tax

Front-loading the carbon tax offers the advantage that growth in the cumulative carbon stock slows earlier, and therefore a lower tax rate can achieve the same climate target. Similarly to how $\bar{\tau}_c^i$ was estimated for $T_{\tau_c}^{BL} = 25$, I find the endogenous tax rate when the tax is implemented in $T_{\tau_c}^{FL} = 15$ time periods for each instrument. In figure 10, I plot the results of the endogenous tax rates when front-loading.

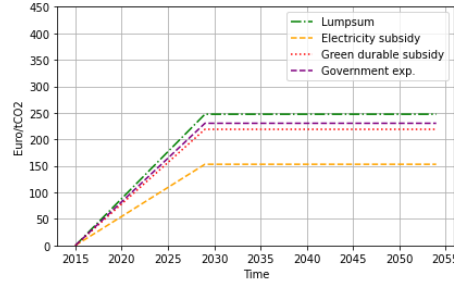


Figure 10: Carbon tax path by revenue-recycling instrument to meet the exogenous temperature target (2 degrees celcius) with front-loading

The qualitative order of tax rates between the instruments remains the same, however compared with the baseline case within instruments, all tax rates are significantly lower in the long run. Therefore, in the new steady state after the transition is complete, GDP and wages are higher than in the baseline case. However, because of front-loading, the carbon tax is higher in the short run than in the baseline case, and since the productivity gains in the electricity sector have not yet fully materialized in the short run (up to $t = 15$), there is a large short-run GDP and wage dip in the front-loading case. This larger short-run dip, combined with a higher fossil fuel price in near-term periods, produce a larger negative aggregate welfare burden for generations alive when the policy is introduced. I visualize this timing trade-off for each instrument in figure 11.

The left panel shows, for each instrument, the GDP decline in the final steady state (x-axis) and the aggregate current welfare loss for alive generations (y-axis) for the baseline (BL) and front-loaded (FL) tax path. The right-hand panel shows the equivalent comparison but with the welfare loss of the lowest income group rather than the aggregate. A downward-sloping curve between FL and BL implies a timing trade-off: increasing the speed of phasing in raises GDP, but increases (decreases) the current welfare loss (gain). The solid lines

show the trade-offs for the case with the electrification margin; the dashed lines show the results when the electrification margin is excluded.

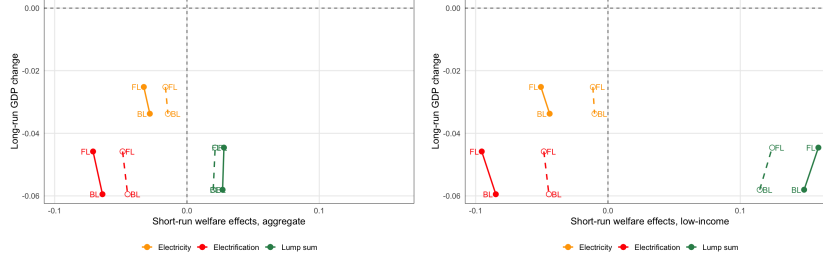


Figure 11: GDP decline in the long run versus aggregate welfare change now (left) and low-income group welfare change now (right) for the baseline (BL) and front-loaded tax (FL) case. Solid line shows results with electrification margin, dashed line shows results without electrification margin.

The figure highlights several results. First, for all instruments, the long run GDP decrease relative to no policy is smaller when front-loading the tax, because of the lower tax rate (figure 10 versus figure 4). a pattern already established above is clearly visible in this figure: the electrification margin affects welfare negatively after a carbon tax is introduced (under non-lump-sum recycling), and this effect is significantly larger for low-income households. Comparing welfare outcomes between front- and back-loading the tax, front-loading increases the magnitude of the welfare effect: for subsidy recycling, welfare becomes more negative, while with lump-sum, it becomes more positive. These welfare effects are larger because of the electrification margin, especially for the poorest households. This implies that front-loading hurts low-income households the most, and the electrification margin is the cause. Because low-income households cannot adopt green technologies early, the higher short-run tax rate hurts them more. Lump-sum redistribution alleviates this timing trade-off.

6 Conclusion

In this paper, I study the effects of a carbon tax on efficiency and inequality in an economy containing both sides of the energy transition: electrification and decarbonization of electricity. I examine these effects under multiple revenue recycling instruments. I show that while carbon taxation robustly accelerates emission reduction, its effectiveness and welfare effects depend heavily on the tax path and the revenue-recycling instrument. I also show that welfare effects are unequally distributed, largely because of the electrification margin.

I show that the electrification margin makes the welfare burden to be larger at the lower-end of the income distribution. When subsidy recycling is used, these effects are salient; they erode when households receive all tax revenues back in the form of a uniform lump-sum transfer. However, this comes at a

cost of a lower GDP in the final steady state, caused by a higher maximum tax rate. Combining front-loading with lump-sum redistribution can offset this cost. Under this combination, front-loading does not decrease aggregate welfare, is more progressive, and produces a smaller GDP decrease than the baseline case.

Overall, the electrification margin for households created distributional effects via heterogeneous carbon intensity of energy demand, that are significant. When policy-makers are intent on reducing emissions, they can best balance this objective with an equity objective through the use of lump-sum redistribution rather than targeted subsidies, and phase the tax in quickly to diminish future costs of the carbon tax. These results underscore that the design of revenue recycling determines who wins and who loses from climate policy. By quantifying the relative importance of the underlying inequality channels, the paper offers guidance for policymakers seeking to balance equity and decarbonization efficiency objectives along the green transition.

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Appendix

A Calibration data

A.1 DNB household survey

The DNB household survey consists of longitudinal data following about 2000 households. The survey contains multiple topics, such as work, pensions, mortgages, income, assets, liabilities and others. I use information about income, assets and liabilities from the AGI (Aggregated data on income) and AGW (Aggregated data on assets, liabilities and mortgages) datasets. These datasets contain information about for example total assets or income, as a sum of individual components that were asked in the surveys. The data I use is from the years 2010 to 2020.

To estimate the income process, I constrain the sample to the ages 25-65. I delete observations with a negative net labor income or net labor income that exceeds 300.000 euros per year, to not include outliers.

For the wealth moments, I treat negative wealth levels as zero-wealth in order to match the models no unsecured borrowing constraint. Net worth is calculated as the sum of checking account, savings, bond holdings, stock holdings and mutual funds, and cars. As I am interested in homeowners and car owners making the investment into a cleaner option, for the wealth moments I only use data on homeowners, but exclude housing wealth from the net worth sum. If I were to include renters, the sum of wealth would be lower and I would underestimate the financial ability of homeowners to make a green investment.

A.2 Heatpump uptake

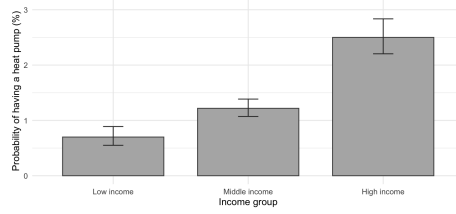


Figure 12: The predicted share of households for different income groups with a heatpump in 2018 using the WOon-dataset from Statistics Netherlands, for homeowners (37750 obs.), mean is 1.41% (dashed line). The correlation persists when controlling for square meters of the house and household age. The correlation is also found for 2021 and 2024.

A.3 EV uptake

B Solving the household problem

The value function is given by

$$V_j(a, \phi, m, y; P, S) = \max\{V_j^{NoAdjust}(a, \phi, m, y; P, S), V_j^{Electrify}(a, \phi, m, y; P, S) + \mu_{Electrify}\} \quad (39)$$

The preference shock μ_{Adj} causes that this extensive margin becomes a switching probability in the expectation, i.e. before realization of the shock. The probability is given by

$$P^{Electrify}(a, \phi, m, y; P, S) = 1 - \exp\left(-\exp\left(\frac{V_j^{Electrify}(a, \phi, m, y; P, S) + \mu_{Electrify} - V_j^{NoAdjust}(a, \phi, m, y; P, S)}{\sigma_{adj}}\right)\right) \quad (40)$$

I individually solve for the intermediate functions, which I can do for the last living period given the bequest value. Then I use backward induction to solve for the remainder of the periods.

I start by solving the non-adjuster problem. I first solve for the static problem analytically. The dynamic problem then becomes

$$V_j^{NA}(a, \phi, m, y; P, S) = \max_{a'} \tilde{u}(\tilde{c}) + \beta \mathbb{E}[V_{j+1}(a', \phi', m', y'; P', S')] \quad (41)$$

subject to

$$\tilde{w} = wy + (1+r)a + \mathcal{T} - \delta_\phi P^{d_\phi} - \pi(j, t)m - a' \quad (42)$$

$$e = (1-\gamma)\frac{\tilde{w}}{p_\phi} + \gamma\frac{\bar{s}}{A_{s,\phi}} \quad (43)$$

$$c = \gamma \left(\tilde{w} - \frac{p_\phi}{A_{s,\phi}} \bar{s} \right) \quad (44)$$

$$\tilde{c} = c^\gamma (A_{s,\phi}e - \bar{s})^{1-\gamma} \quad (45)$$

$$\tilde{u}(\tilde{c}) = \frac{\tilde{c}^{1-\sigma}}{1-\sigma} \quad (46)$$

$$m' = (1+r+r^m - \pi(j, t))m, \quad \phi' = \phi \quad (47)$$

$$a' \geq 0 \quad (48)$$

where $A_{s,\phi} \equiv \phi A_{s,G} + (1-\phi)A_{s,B}$ and $p_\phi \equiv \phi P^E + (1-\phi)(P^{FF} + \tau_c)$. I solve the non-adjuster problem using the endogenous grid method with upper envelope as described in Druedahl (2021).

Given value function $V_j^{NA}(a, \phi, m, y; P, S)$ and policy functions $a'_j(a, \phi, m, y; P, S)$, $c_j(a, \phi, m, y; P, S)$, $e_j(a, \phi, m, y; P, S)$, I continue by solving for the electrifier value function, by recognizing that this is an interpolation of the non-adjuster problem

$$V_j^{Electrify}(a, \phi, m, y; P, S) = \max_{m'} V_j^{NA}(\tilde{a}, \phi' = 1, m', y; P, S) \quad (49)$$

$$\tilde{a} = a - (P^G(1 - s_{dG}) - \varphi P^B) + m' \quad (50)$$

$$m' \leq \lambda(1 - s_{dG})P^G \quad (51)$$

Which gives policy function $\phi'_j(a, \phi, m, y; P, S)$, $m'_j(a, \phi, m, y; P, S)$ and value function $V_j^{Electrify}(a, \phi, m, y; P, S)$

C Solving the Production side

There are three production sectors. One for the final output good, one for the two durable goods and one for electricity.

C.1 Output-good-producing firm

A representative firm produces output goods which can be used for production or converted in durable goods. Inputs are capital, labor and electricity. The technology is a constant returns to scale CES

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1-\psi)(A_{KC}K_C^{\alpha_c}L_C^{1-\alpha_c})^{\frac{\nu-1}{\nu}} + \psi A_{MC}M^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}} \quad (52)$$

Gives first order conditions

$$r + \delta = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1 - \psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} \alpha_c K_C^{\alpha_c-1} L_C^{1-\alpha_c} \quad (53)$$

$$w = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1 - \psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} (1 - \alpha_c) K_C^{\alpha_c} L_C^{-\alpha_c} \quad (54)$$

$$P^E = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1} \right) \quad (55)$$

$$P^{FF} + \tau_c = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1 - \kappa) F F_C^{\frac{\iota-1}{\iota}-1} \right) \quad (56)$$

where

$$B = A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c} \quad (57)$$

$$M = \left(\kappa E_C^{\frac{\iota-1}{\iota}} + (1 - \kappa) F F_C^{\frac{\iota-1}{\iota}} \right)^{\frac{\iota}{\iota-1}}. \quad (58)$$

C.2 Durable good producers

In this economy, the durable goods can be produced by converting the consumption good (with cost 1, as its the numeraire) at some rate A_G, A_B , such that production functions are given as

$$Y^{D_G}(C_G) = A_G C_G \quad (59)$$

$$Y^{D_B}(C_B) = A_B C_B \quad (60)$$

These imply input demand functions

$$C_G = \frac{1}{A_G} Y^{D_G} \quad (61)$$

$$C_B = \frac{1}{A_B} Y^{D_B} \quad (62)$$

and prices $P^G = \frac{1}{A_G}$, $P^B = \frac{1}{A_B}$

C.3 Electricity firm

The profits of the electricity firm are given as follows

$$\begin{aligned} \Pi(K_{E_R}, K_{E_F}, FF_E) = & P_E \left(\alpha_R^{\frac{1}{\lambda}} (A_R K_R)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} (A_F [\zeta K_F^{\frac{\omega-1}{\omega}} + (1 - \zeta) F F^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}})^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} \\ & - (r + \delta) K_{E_R} - (r + \delta) K_{E_F} - (P^{FF} + \tau_c) FF_E \end{aligned} \quad (63)$$

Equivalently, I can write the problem as follows

$$\Pi = P_E \left(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} - P_{E_R} E_R - P_{E_F} E_F \quad (64)$$

subject to the technology constraints

$$E_R = A_R K_R \quad (65)$$

$$E_F = A_F [\zeta^{\frac{1}{\omega}} (A_{KF} K_F)^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} F F_E^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}} \quad (66)$$

Which implies I can split up the problem in two parts. First the problem where I minimize cost by choosing K_{E_R} ($K_{E_R}, F F_E$) subject to the constraint that production needs to equal some amount E_R (E_F). This problem gives me input demand functions for K_{E_R} ($K_{E_R}, F F_E$) in terms of E_R (E_F) as well as a price for E_R (E_F), namely P_{E_R} (P_{E_F}).

With these prices I can solve the problem in equation 64, to get the input demand functions for E_F and E_R in terms of some output level Y^E . Combining these with the input functions derived from the other two problems, I get input demand functions: $K_{E_R}(Y^E, r, P^F F, \tau_c)$, $K_{E_R}(Y^E, r, P^F F, \tau_c)$, $F F_E(Y^E, r, P^F F, \tau_c)$, as functions of total output and input prices.

The first step to solving the firm problem for the supply function, is to choose E_F, E_R to minimize cost subject to the constraint of producing some level of electricity $\bar{E}$. The Lagrangian reads

$$\mathcal{L} = P_{E_R} E_R + P_{E_F} E_F + \Lambda (\bar{E} - (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}})$$

The first order conditions are

$$\frac{d\mathcal{L}}{dE_R} \implies \Lambda \alpha_R^{\frac{1}{\lambda}} (E_R)^{\frac{\lambda-1}{\lambda}-1} (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}-1} = P_{E_R} \quad (67)$$

$$\frac{d\mathcal{L}}{dE_F} \implies \Lambda \alpha_F^{\frac{1}{\lambda}} (E_F)^{\frac{\lambda-1}{\lambda}-1} (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}-1} = P_{E_F} \quad (68)$$

$$\frac{d\mathcal{L}}{d\Lambda} \implies \bar{E} = (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}} \quad (69)$$

Dividing through

$$\frac{P_{E_R}}{P_{E_F}} = \frac{\alpha_R^{\frac{1}{\lambda}} (E_R)^{\frac{\lambda-1}{\lambda}-1}}{\alpha_F^{\frac{1}{\lambda}} (E_F)^{\frac{\lambda-1}{\lambda}-1}} \quad (70)$$

Solve for E_R and plug in

$$\left(\frac{P_{E_R}}{P_{E_F}} \right)^{-\lambda} \frac{\alpha_R}{\alpha_F} E_F = E_R \implies \bar{E} = (\alpha_R^{\frac{1}{\lambda}} \left(\frac{P_{E_R}}{P_{E_F}} \right)^{-\lambda} \frac{\alpha_R}{\alpha_F} E_F)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}}$$

Rewriting:

$$\bar{E} = \left(\alpha_R^{\frac{1}{\lambda}} \left(\frac{P_{E_R}}{P_{E_F}} \right)^{-\lambda} \frac{\alpha_R}{\alpha_F} \right)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} E_F \quad (71)$$

such that

$$E_F = \frac{\bar{E}}{\left(\alpha_R^{\frac{1}{\lambda}} \left(\frac{P_{E_R}}{P_{E_F}} \right)^{-\lambda} \frac{\alpha_R}{\alpha_F} \right)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}}} \quad (72)$$

where input demand scales linearly in the amount produced $\bar{E}$, taking all prices as given.

This implies that for any output $\bar{E}$, input demand is given by

$$E_F = \alpha_F \left(\frac{P_{E_F}}{P_E} \right)^{-\lambda} \bar{E} \quad (73)$$

$$E_R = \alpha_R \left(\frac{P_{E_R}}{P_E} \right)^{-\lambda} \bar{E} \quad (74)$$

and the natural price index for E is

$$P_E = (\alpha_R P_{E_R}^{1-\lambda} + \alpha_F P_{E_F}^{1-\lambda})^{\frac{1}{1-\lambda}} \quad (75)$$

Similarly, for the intermediate electricity from renewable and fossil, from the first order conditions, prices for these are given by marginal costs

$$P_{E_R} = \frac{r + \delta}{A_R} \quad (76)$$

$$P_{E_F} = \frac{1}{A_F} (\zeta((r + \delta)/A_{KF})^{1-\omega} + (1 - \zeta)(P_{FF} + \tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \quad (77)$$

and input demands are

$$K_R = \frac{1}{A_R} E_R \quad (78)$$

$$K_F = \zeta \left(\frac{r + \delta}{A_{KF} P_{E_F}} \right)^{-\omega} E_F \quad (79)$$

$$FF_E = (1 - \zeta) \left(\frac{P_{FF} + \tau_c}{P_{E_F}} \right)^{-\omega} E_F \quad (80)$$

D Equilibrium algorithm

D.1 No tax, stationary equilibrium

Given exogenous fossil fuel price P^{FF}

1. Initialization.

- (a) Guess an interest rate r
- (b) Compute the implied price of electricity and the prices of durables as marginal costs:

$$P^E = \left[\alpha_R \left(\frac{r}{A_R} \right)^{1-\lambda} + \alpha_F \left(\frac{1}{A_F} (\zeta r^{1-\omega} + (1-\zeta)(P_{FF} + \tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \right)^{1-\lambda} \right]^{\frac{1}{1-\lambda}} \quad (81)$$

$$P^G = \frac{1}{A_G}, \quad P^B = \frac{1}{A_B} \quad (82)$$

2. Firm Problem.

- (a) Using a non-linear solver, solve for (K_C, E_C, FF_C, w) such that the labor market clears and the firm's first-order conditions are satisfied:

$$r + \delta = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1-\psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} \alpha_c K_C^{\alpha_c-1} L_C^{1-\alpha_c} \quad (83)$$

$$w = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1-\psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} (1-\alpha_c) K_C^{\alpha_c} L_C^{-\alpha_c} \quad (84)$$

$$P^E = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1} \right) \quad (85)$$

$$P^{FF} + \tau_c = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1-\kappa) FF_C^{\frac{\iota-1}{\iota}-1} \right) \quad (86)$$

3. Household Problem.

- (a) Given the price vector $P = (P^{FF}, P^G, P^B, P^E, r, w)$, compute the value,

$$V_j^{NA}(a, \phi, m, y; P, S), V_j^A(a, \phi, m, y; P, S),$$

and policy functions,

$$\phi'_j(a, \phi, m, y; P, S), m'_j(a, \phi, m, y; P, S), a'_j(a, \phi, m, y; P, S), \\ c_j(a, \phi, m, y; P, S), e_j(a, \phi, m, y; P, S).$$

- (b) Compute the stationary distribution μ such that the bequest condition holds:

$$\int a'(\mathbf{x}) d\mu_J = \int \frac{a}{1+r}, d\mu_{-1} \quad (87)$$

4. Aggregation.

(a) Compute household aggregates:

$$A_{hh} = \sum_{j=0}^{J-1} \int a(\mathbf{x}), d\mu_j, \quad C_{hh} = \sum_{j=0}^{J-1} \int c, d\mu_j, \quad (88)$$

$$E_{hh} = \sum_{j=0}^{J-1} \int \phi' e, d\mu_j, \quad FF_{hh} = \sum_{j=0}^{J-1} \int (1 - \phi') e, d\mu_j \quad (89)$$

(b) Compute net durable demands:

$$D_{hh}^G = \sum_{j=0}^{J-1} \int [\phi \delta_G + (\phi' - \phi)], d\mu_j \quad (90)$$

$$D_{hh}^B = \sum_{j=0}^{J-1} \int [(1 - \phi) \delta_B - \varphi(\phi' - \phi)], d\mu_j \quad (91)$$

5. Sectoral Output and Input Demands.

(a) Use market clearing for electricity:

$$Y^E = E_{hh} + E_C \quad (92)$$

(b) Compute input demands for electricity production:

$$K_{E_R} = \frac{1}{A_R} \alpha_R \left(\frac{P^{E_R}}{P^E} \right)^{-\lambda} Y^E, \quad (93)$$

$$K_{E_F} = \zeta \left(\frac{r}{P^{E_F}} \right)^{-\omega} \alpha_F \left(\frac{P^{E_F}}{P^E} \right)^{-\lambda} Y^E \quad (94)$$

$$FF_E = (1 - \zeta) \left(\frac{P^{FF} + \tau_c}{P^{E_F}} \right)^{-\omega} \alpha_F \left(\frac{P^{E_F}}{P^E} \right)^{-\lambda} Y^E \quad (95)$$

(c) Prices of intermediate inputs:

$$P_{E_R} = \frac{r}{A_R}, \quad P_{E_F} = \frac{1}{A_F} (\zeta^\omega r^{1-\omega} + (1 - \zeta)^\omega (P_{FF} + \tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \quad (96)$$

6. Durable Goods Sectors.

(a) Use market clearing for durables:

$$Y^{D_G} = D_{hh}^G, \quad Y^{D_B} = D_{hh}^B \quad (97)$$

(b) Compute input demands:

$$C_G = \frac{Y^{D_G}}{A_G}, \quad C_B = \frac{Y^{D_B}}{A_B} \quad (98)$$

7. Equilibrium Conditions.

- (a) Check capital market clearing:

$$A = K_C + K_{E_R} + K_{E_F} \quad (99)$$

- (b) Check aggregate resource constraint:

$$Y = C_{hh} + C_B + C_G + I_{E_R} + I_{E_F} + I_C + P^{FF}(FF_{hh} + F_E) + \int \phi f(j)M - (\phi' - \phi)M', d\mu \quad (100)$$

If either condition fails, update r and return to Step 1(b).

D.2 Transition

1. Initialization.

- (a) Fix a time horizon $t = 0, \dots, T$, tolerance levels $\varepsilon_r > 0$, and a damping parameter $\eta_r \in (0, 1)$.
- (b) Choose an initial guess for the interest-rate path $\{r_t^{(i=0)}\}_{t=0}^T$.
- (c) Specify path for productivity gains $\{A_t\}_{t=0}^T$ and taxes $\{\tau_{ct}\}_{t=0}^T$, and calculate the implied sequences for $\{P_t^E\}_{t=0}^T$ and $\{w_t\}_{t=0}^T$ from the firm problem.

2. Outer iteration $i = 0, 1, 2, \dots$ (until convergence):

- (a) Compute the value function of the stationary equilibrium in the final period T , $V_j^T(a, \phi; Y, P, \mathcal{T})$. By backward induction, taking the price sequences as given, compute the value function $V_j^t(a, \phi; Y, P, \mathcal{T})$ for $t = T - 1, \dots, 0$. Using the policy functions, simulate the distribution forward for T periods starting from the initial distribution μ_0 , being the initial cross-sectional distribution of households μ_0 as the steady state at $t = 0$.
- (b) From the distribution, for each time period t calculate aggregate capital holdings $\int a(r_i)d\mu_t$ and aggregate electricity demand from households $E_{hh} = \int \phi ed\mu_t$.
- (c) **Inner iteration**, $j = 0, 1, 2, \dots$ (until convergence): solve the static production problem to clear the market. For each time period t :
 - i. Guess an interest rate r_j , calculate $P_j^E(r_j)$.
 - ii. Given r_j , $P_j^E(r_j)$ calculate K_C and E_C from the first order conditions of the final consumption good firm.
 - iii. Given E_{hh} , E_C and r_j calculate K^{E_R} and K_{E_F} .
 - iv. Check if $\int a(r_i)d\mu_t = K_C(r_j) + K^{E_R}(r_j) + K_{E_F}(r_j)$. If not, update r_j and go back to ii.

(d) Check if $\{r_t^{(i)}\}_{t=0}^T = \{r_t^{(j)}\}_{t=0}^T$, if not update

$$\{r_t^{(i+1)}\}_{t=0}^T = \eta_r \{r_t^{(i)}\}_{t=0}^T + (1 - \eta_r) \{r_t^{(j)}\}_{t=0}^T$$

and continue with step 1.c.

E Deriving the aggregate resource constraint

Given production functions for durable goods, consumption and electricity

$$D_G : Y^{d_G}(C_G) = A_G C_G \quad (101)$$

$$D_B : Y^{d_B}(C_B) = A_B C_B \quad (102)$$

$$C : Y^C(K_C, L_C, E_C, FF_C) = F^C(K_C, L_C, E_C, FF_C) \quad (103)$$

$$E : Y^E(K_{E_F}, K_{E_R}, FF_E) = F^E(K_{E_F}, K_{E_R}, FF_E) \quad (104)$$

and the fact that fossil fuels are imported, the following markets need to clear²²:

$$\text{Capital:} \quad \sum_{j=1}^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \quad (105)$$

$$\text{Labor:} \quad \sum_{j=1}^{J-1} \int y d\mu_j = L_C (= \bar{L}) \quad (106)$$

$$\text{Green Durable:} \quad Y^{D_G} = \sum_{j=1}^{J-1} \int [(\phi' - \phi) + \delta_G \phi] d\mu_j \quad (107)$$

$$\text{Brown Durable:} \quad Y^{D_B} = \sum_{j=1}^{J-1} \int [\delta_B(1 - \phi) - \varphi(\phi' - \phi)] d\mu_j \quad (108)$$

$$\text{Fossil Fuel:} \quad Y^{FF} = FF_{HH} + FF_E + FF_C \quad (109)$$

$$\text{Electricity:} \quad Y^E = \sum_{j=1}^{J-1} \int e \phi' d\mu_j + E_C \quad (110)$$

the government budget constraint needs to hold

$$G + \mathcal{T} + s_{ER} K_{ER}(r + \delta) + s_{dG} P^G \int (\phi' - \phi) d\mu = \tau_c [FF_{HH} + FF_E + FF_C] \quad (111)$$

such that the aggregate resource constraint can be derived. Without loss of generality, I here only derive the case for lump-sum redistribution.

²²The conditions read as follows. Total saving must equal total capital demand. Total labor supply must equal labor demand. Production of the green durable must be equal to newly bought green durables plus depreciation (bequests don't go on the market but are passed through as the durable itself), production of brown durable equals depreciation minus what comes on the market (can be negative?). Total fossil fuel equals household usage and fossil fuels used in electricity production, lastly electricity production is equal to household demand plus final consumption good electricity input demand

Starting from the household budget constraint (note that households with $\phi = 1$ are not allowed choose to switch back after electrification, such that the budget constraint can be written as):

$$wy + (1+r)a + \mathcal{T} = c + a' + \phi' e P^E + (1 - \phi') e (P^{FF} + \tau_c) + \phi \delta_G P^{d_G} + (1 - \phi) \delta_B P^{d_B} + (\phi' - \phi)(P^{d_G} - \varphi P^{d_B}) + \underbrace{\phi \pi(j)m - (\phi' - \phi)m'}_{=m(\phi', \phi)} \quad (112)$$

where $(\phi' - \phi)(P^{d_G} - \varphi P^{d_B})$ is the cost of electrification. Furthermore $m(\phi', \phi)$ is the loan installment payment in case of staying (having previously electrified) or receiving the loan in case of electrifying.

In the first step I integrate both sides of the budget constraint

$$\begin{aligned} wL_C + \int (1+r)a d\mu + \mathcal{T} = \\ C_{hh} + \int a' d\mu + \int \phi' e P^E d\mu + \int (1 - \phi') e (P^{FF} + \tau_c) d\mu + \\ \int \phi \delta_G P^{d_G} d\mu + \int (1 - \phi) \delta_B P^{d_B} d\mu + \\ \int (\phi' - \phi)(P^{d_G} - \varphi P^{d_B}) d\mu + \int m(\phi', \phi) d\mu \end{aligned} \quad (113)$$

Starting with the the two durable goods, in the budget constraint, rearrange terms to get these:

$$P^{d_B} \left[\int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu \right] \quad \text{and} \quad P^{d_G} \left[\int \phi \delta_G + (\phi' - \phi) d\mu \right] \quad (114)$$

which are exactly as in market clearing conditions (7) and (8), such that I can write

$$P^{D_B} \left[\int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu \right] = P^{D_B} Y^{D_B} \quad \text{and} \quad P^{D_G} \left[\int \phi \delta_G + (\phi' - \phi) d\mu \right] = P^{D_G} Y^{D_G} \quad (115)$$

Furthermore, plugging the production functions for $Y^{d_B} = A_G C_G$ and $Y^{d_G} = A_G C_G$, and noting that in this linear technology, and with price for C equal to one, I get $P^{d_G} = \frac{1}{A_G}$ and $P^{d_B} = \frac{1}{A_B}$ these expressions simply become

$$P^{D_B} \left[\int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu \right] = C_B \quad \text{and} \quad P^{D_G} \left[\int \phi \delta_G + (\phi' - \phi) d\mu \right] = C_G \quad (116)$$

which I can substitute in equation 113 to get

$$wL_C + \int (1+r)a d\mu + \mathcal{T} = C_{hh} + C_B + C_G + \int a' d\mu + \int \phi' e P^E d\mu + \int (1-\phi') e (P^{FF} + \tau_c) d\mu + \int m(\phi', \phi) d\mu \quad (117)$$

Next, I rewrite the market clearing condition for electricity,

$$\sum_{j=1}^{J-1} \int e \phi' d\mu_j = Y^E - E_C \quad (118)$$

and substitute the RHS in the new aggregate budget constraint. I also take $(1+r)a$ to the right hand side, to get

$$wL_C + \mathcal{T} = C_{hh} + C_B + C_G + \int [a' - (1+r)a] d\mu + (Y^E - E_C) P^E + \int (1-\phi') e (P^{FF} + \tau_c) d\mu + \int m(\phi', \phi) d\mu \quad (119)$$

Using the capital market clearing condition, I rewrite $\int [a' - (1+r)a] d\mu$ as follows:

$$[a' - (1+r)a] d\mu = K_{t+1} - (1+r)K_t \quad (120)$$

Noting that $K_t = K_{C,t} + K_{E_R,t} + K_{E_F,t}$, I split up aggregate capital to get:

$$K_{t+1} - (1+r)K_t = K_{C,t+1} - K_{C,t} + K_{E_R,t+1} - K_{E_R,t} + K_{E_F,t+1} - K_{E_F,t} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} \quad (121)$$

$$= I_{C,t} + I_{E_R,t} + I_{E_F,t} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} \quad (122)$$

which I substitute in the aggregate budget constraint

$$wL_C + \mathcal{T} = C_{hh} + C_B + C_G + I_C + I_{E_R} + I_{E_F} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} + (Y^E - E_C) P^E + \int (1-\phi') e (P^{FF} + \tau_c) d\mu + \int m(\phi', \phi) d\mu \quad (123)$$

Take $-rK_{C,t}$ and $-P^E E_C$ to the other side to get

$$wL_C + rK_{C,t} + P^E E_C + \mathcal{T} = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} - rK_{E_R} - rK_{E_F} + Y^E P^E + \int (1-\phi') e (P^{FF} + \tau_c) d\mu + \int m(\phi', \phi) d\mu \quad (124)$$

and since standard FOC for the final production good imply that marginal products are equalized with input prices, Eulers theorem implies that $Y^C = wL_C + (\delta + r)K_{C,t} + P^E E_C + (P^{FF} + \tau_c)FF_C$:

$$Y^C + \mathcal{T} = (P^{FF} + \tau_c)FF_C + C_{hh} + C_B + C_G + I_C + I_{E_R} + I_{E_F} - rK_{E_R} - rK_{E_F} + Y^E P^E + \delta K_C \quad (125)$$

$$+ \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu$$

There is a zero-profit condition in the market for electricity which reads that

$$Y^E P^E - (\delta + r)K_{E_R,t} - (\delta + r)K_{E_F,t} - FF_E(P^{FF} + \tau_c) = 0 \quad (126)$$

So I can substitute for $Y^E P^E - rK_{E_R,t} - rK_{E_F,t}$ in the aggregate budget constraint

$$Y^C + \mathcal{T} = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + (P^{FF} + \tau_c)FF_C + FF_E(P^{FF} + \tau_c) \quad (127)$$

$$+ \delta(K_C + K_{ER} + K_{EF}) + \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu$$

Next I impose the government budget constraint such that $\mathcal{T}$ cancels with $\tau_c \int e(1 - \phi')d\mu + \tau_c FF_E + \tau_c FF_C$

$$Y^C = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + P^{FF}FF_C + FF_E P^{FF} \quad (128)$$

$$+ \delta(K_C + K_{ER} + K_{EF}) + P^{FF} \int (1 - \phi')ed\mu + \int m(\phi', \phi)d\mu$$

Impose market clearing for fossil fuels (with abroad) $FF = FF_C + FF_E + \int (1 - \phi')ed\mu$

$$Y^C = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + P^{FF}FF + \int m(\phi', \phi)d\mu + \delta(K_C + K_{ER} + K_{EF}) \quad (129)$$

Substitute in

$$\int m(\phi', \phi)d\mu = \int \phi f(j)M - (\phi' - \phi)M'd\mu \quad (130)$$

And rewrite capital investment

$$Y_t^C = C_{hh,t} + C_{B,t} + C_{G,t} + K_{t+1} - K_t + P^{FF}FF_t + \int \phi f(j)M_t - (\phi' - \phi)M_{t+1}]d\mu \quad (131)$$

Which reads as follows. Total production can either be used for consumption of the household directly, can be converted in brown or durable goods, can be invested in capital, can be spent on fossil fuels or spend on repaying the loan. In steady state,

$$Y^C = C_{hh} + C_B + C_G + \delta K + P^{FF}FF + M(r + r_m) \quad (132)$$

F Climate Model

F.1 The Climate Model

The climate module is exactly taken from Douenne et al. (2025). The temperature is given as

$$Z_{t+1} = Z_t + \epsilon (\zeta \mathcal{E}_t - Z_t), \quad (133)$$

where cumulative emissions evolve according to

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}. \quad (134)$$

I set Z_0 , ϵ , ζ , $\mathcal{E}_0$, and the exogenous emissions path $\{E_t^{\text{ex}}\}_{t \geq 0}$ externally, and take all values from Douenne et al. (2025).

F.2 Mapping model outcomes to emissions

In the model, I explicitly track fossil fuel demand, FF_t , as the sum of direct household and firm fossil fuel demand and indirect fossil fuel demand in electricity production:

$$FF_t = FF_{E,t} + FF_{C,t} + FF_{HH,t}. \quad (135)$$

The exogenous price of fossil fuels is calibrated as the price of 1000 kWh of energy, where 67% stems from gasoline and 33% from natural gas. The 670 kWh gasoline component corresponds to

$$\frac{670}{8.9} = 75.2 \text{ liters of gasoline}, \quad (136)$$

and the 330 kWh natural gas component corresponds to

$$\frac{330}{10} = 33 \text{ m}^3 \text{ of natural gas}. \quad (137)$$

Using emission factors of 2.3 kg CO₂ per liter of gasoline and 1.9 kg CO₂ per m³ of natural gas, one unit of fossil fuel in the model produces

$$75.2 \cdot 2.3 + 33 \cdot 1.9 = 235.66 \text{ kg CO}_2. \quad (138)$$

Hence, model emissions implied by fossil fuel demand are given by

$$E_t^M = 235.66 \cdot FF_t, \quad (139)$$

with E_t^M measured in kg CO₂.

F.3 Temperature outcomes

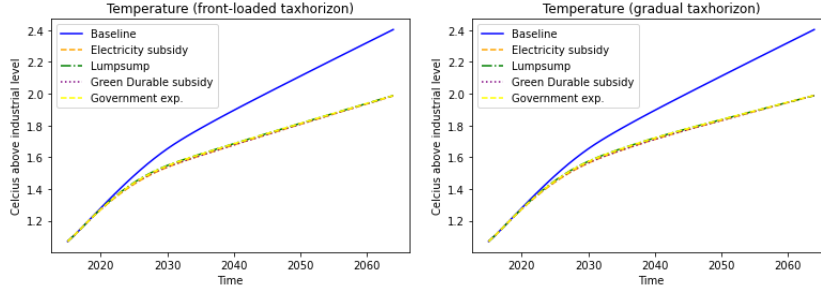


Figure 13: Temperature paths by implementation horizon and revenue-recycling instrument in order to meet the exogenous temperature target (2 degrees celcius)

G Lifetime Consumption Decomposition

I derive the decomposition of lifetime consumption. In doing so, I express the total expected lifetime consumption (plus savings bequests) of a household as the sum of its lifetime income sources, net of energy and durable expenditures.

Step 1: Per-period budget constraint. At age j and time t , the household's budget constraint is:

$$c_{j,t} + P^{\phi_{j,t}} e_{j,t} + a'_{j,t} = \mathcal{T}_t + w_t y_j + (1+r_t) a_{j,t} - \delta_{\phi_{j,t}} - f(j) m_{j,t} + (\phi'_{j,t} - \phi_{j,t}) [m'_{j,t} - P_t^G + \varphi P^B] \quad (140)$$

where the left-hand side contains consumption, energy expenditure, and savings, and the right-hand side contains transfers, labor income, asset income, maintenance costs, loan repayments, and the net cost of switching to a green durable. To keep notation compact, define:

$$EE_{j,t} \equiv P^{\phi_{j,t}} e_{j,t} \quad (141)$$

$$DE_{j,t} \equiv \delta_{\phi_{j,t}} + f(j) m_{j,t} - (\phi'_{j,t} - \phi_{j,t}) [m'_{j,t} - P_t^G + \varphi P^B] \quad (142)$$

as energy expenditure and net durable expenditure respectively. The budget constraint then simplifies to:

$$c_{j,t} = \mathcal{T}_t + w_t y_j + (1+r_t) a_{j,t} - a'_{j,t} - EE_{j,t} - DE_{j,t} \quad (143)$$

Step 2: Summing over the remaining lifetime. Consider a household of age j_0 at the time the policy is introduced, $t = 0$. This household has $S \equiv J - 1 - j_0$ remaining periods of life. Summing equation (143) over $s = 0, 1, \dots, S$,

where time $t = s$ and the household's age at time s is $j_0 + s$:

$$E_0 \sum_{s=0}^S c_{j_0+s, s} = E_0 \sum_{s=0}^S [\mathcal{T}_s + w_s y_{j_0+s} + (1 + r_s) a_{j_0+s, s} - a'_{j_0+s, s} - \text{EE}_{j_0+s, s} - \text{DE}_{j_0+s, s}] \quad (144)$$

Step 3: Telescoping the asset terms. Using the fact that $a'_{j,t} = a_{j+1, t+1}$, the asset terms form a telescoping sum. Writing them out explicitly:

$$\begin{aligned} E_0 \sum_{s=0}^S [(1 + r_s) a_{j_0+s, s} - a'_{j_0+s, s}] &= E_0 \left[[(1 + r_0) a_{j_0, 0} - a_{j_0+1, 1}] \right. \\ &\quad + [(1 + r_1) a_{j_0+1, 1} - a_{j_0+2, 2}] \\ &\quad + \dots \\ &\quad \left. + [(1 + r_S) a_{j_0+S, S} - a_{j_0+S+1, S+1}] \right] \end{aligned}$$

For each $s \geq 1$, the asset $a_{j_0+s, s}$ appears as $-a_{j_0+s, s}$ from the previous row and $+(1 + r_s) a_{j_0+s, s}$ from its own row, netting to $+r_s \cdot a_{j_0+s, s}$. The final term $a_{j_0+S+1, S+1} = a_{J, T}$ is the stock of assets left at death, which I denote as the bequest a_J . The telescoping sum therefore equals:

$$E_0 \sum_{s=0}^S [(1 + r_s) a_{j_0+s, s} - a'_{j_0+s, s}] = (1 + r_0) a_{j_0, 0} + E_0 \left[\sum_{s=1}^S r_s a_{j_0+s, s} - a_J \right] \quad (145)$$

Step 4: Lifetime consumption identity. Substituting (145) back into (144) and moving the bequest a_J to the left-hand side:

$$\underbrace{E_0 \sum_{s=0}^S c_{j_0+s, s} + a_J}_{\text{Lifetime Consumption}} = E_0 \left[\underbrace{(1 + r_0) a_{j_0, 0}}_{\text{initial wealth}} + \underbrace{\sum_{s=0}^S w_s y_{j_0+s}}_{\text{labor income}} + \underbrace{\sum_{s=1}^S r_s a_{j_0+s, s}}_{\text{asset income}} + \underbrace{\sum_{s=0}^S \mathcal{T}_s}_{\text{transfers}} - \underbrace{\sum_{s=0}^S \text{EE}_{j_0+s, s}}_{\text{energy exp.}} - \underbrace{\sum_{s=0}^S \text{DE}_{j_0+s, s}}_{\text{durable exp.}} \right] \quad (146)$$

This identity states that total lifetime consumption (including bequests) equals initial wealth plus the present value of all income flows, net of energy and durable expenditures. Initial wealth $(1 + r_0) a_{j_0, 0}$ is identical across the policy and no-policy case at $t = 0$, since the policy is introduced unexpectedly.

Step 5: Change in lifetime consumption. Taking the difference $\Delta(\cdot)$ between the policy scenario and the no-policy baseline, and noting that initial

wealth cancels:

$$\begin{aligned} \Delta E_0 \left[\sum_{s=0}^S c_{j_0+s, s} + a_J \right] &= \Delta \sum_{s=0}^S \mathcal{T}_s + \Delta E_0 \sum_{s=0}^S w_s y_{j_0+s} + \Delta E_0 \sum_{s=1}^S r_s a_{j_0+s, s} \\ &\quad - \Delta E_0 \sum_{s=0}^S \text{EE}_{j_0+s, s} - \Delta E_0 \sum_{s=0}^S \text{DE}_{j_0+s, s} \end{aligned} \quad (147)$$

which we write compactly as:

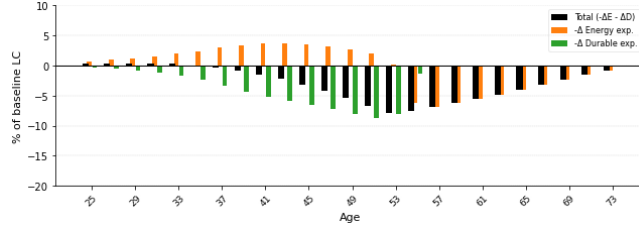
$$\Delta \text{LC} = \Delta \text{Transfers} + \Delta \text{Labor income} + \Delta \text{Asset income} - \Delta \text{Energy expenditure} - \Delta \text{Durable expenditure} \quad (148)$$

where ΔLC denotes the change in lifetime consumption relative to the baseline. A positive value of ΔLC means the policy increases the household's total lifetime consumption capacity. This decomposition allows us to isolate the channels through which climate policy affects household welfare: general equilibrium effects on wages and asset returns, direct redistribution through transfers, and expenditure effects arising from the electrification margin.

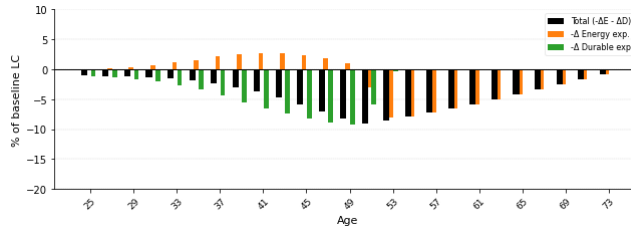
H Extra graphs

H.1 Heterogeneous expenditure changes

Figure 10 for durable subsidy and government expenditure



(a) Durable subsidy



(b) Government expenditure

Figure 14: Heterogeneous expenditure responses across the age space for $z = 0$.

H.2 GDP and prices for combinations of lump-sum and electricity subsidy

First the aggregate outcomes,

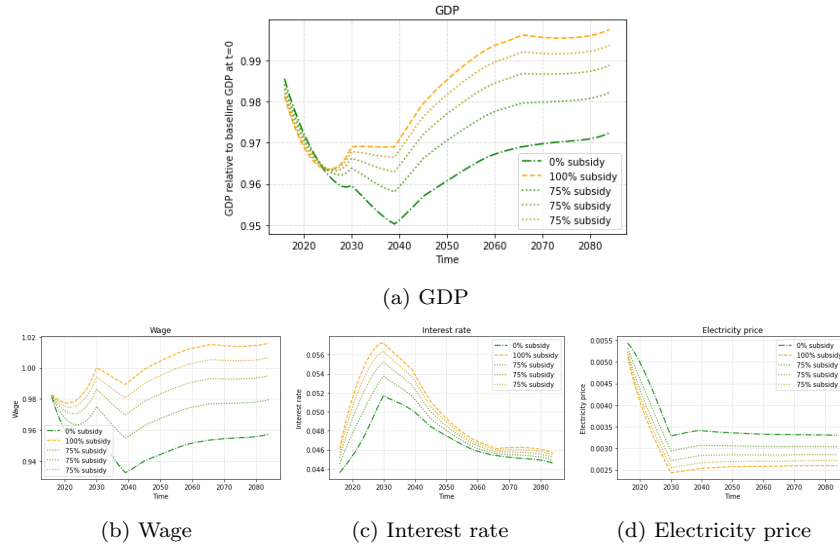


Figure 15: GDP (top) and prices (bottom) over the course of the transition for both subsidies and a combination of the two subsidies

the policies, lump-sum redistribution and tax rate

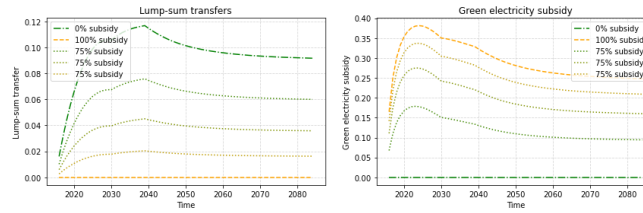


Figure 16: Policies. Right: Lump-sum, left: subsidy

and welfare outcomes

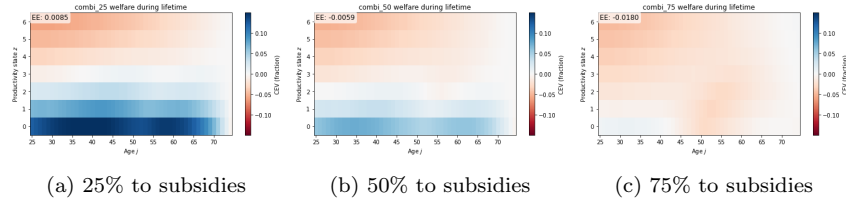


Figure 17: Caption

H.3 GDP and prices for subsidy combinations

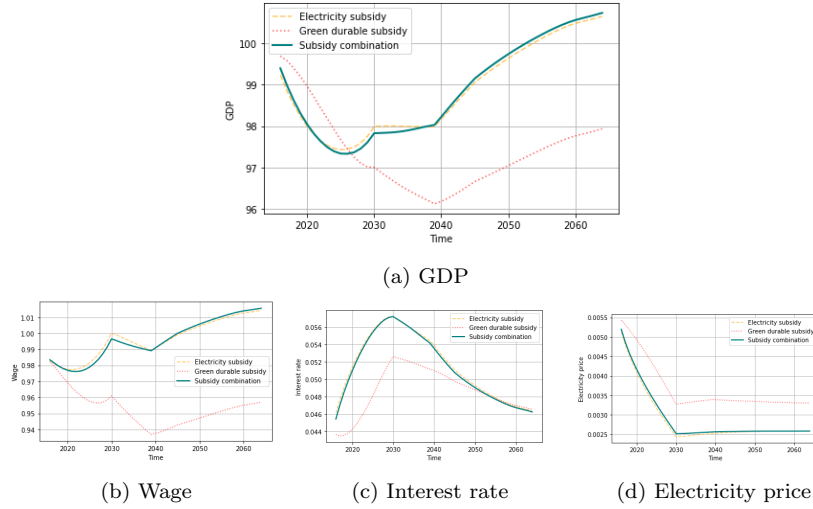


Figure 18: GDP (top) and prices (bottom) over the course of the transition for both subsidies and a combination of the two subsidies

I Robustness

Idiosyncratic income risk A natural source of concern arises from the fact that the lump-sum redistribution scheme alleviates existing distortions, such as idiosyncratic risk and inequalities and is therefore welfare enhancing. In order to test whether the result found above holds true under stricter assumptions, I solve the model for an economy with the same income states but without any risk, i.e. the transition matrix is now an identity matrix, holding everything else constant and re-analyze the welfare results. I find that both the aggregate as well as the distributional incidence of welfare losses remain qualitatively and quantitatively within the same magnitude under the assumption of no idiosyncratic risk. The aggregate gain in welfare from lump-sum distribution reduces from 2.7% to 2.44%. This still does not speak to the concern that lump-sum might increase

aggregate welfare as it alleviates current inequalities, even absent distortions such as risk. However, the conclusion I can draw, and the benefit to lump-sum that is clear compared to all other instruments is that lump-sum is the only instrument which is not regressive. That is, in all cases other than lump-sum the electrification margin causes poor households to suffer the greatest welfare losses as explained above. Lump-sum recycling is the only instrument that alleviates this welfare burden caused by the electrification margin for the poor.

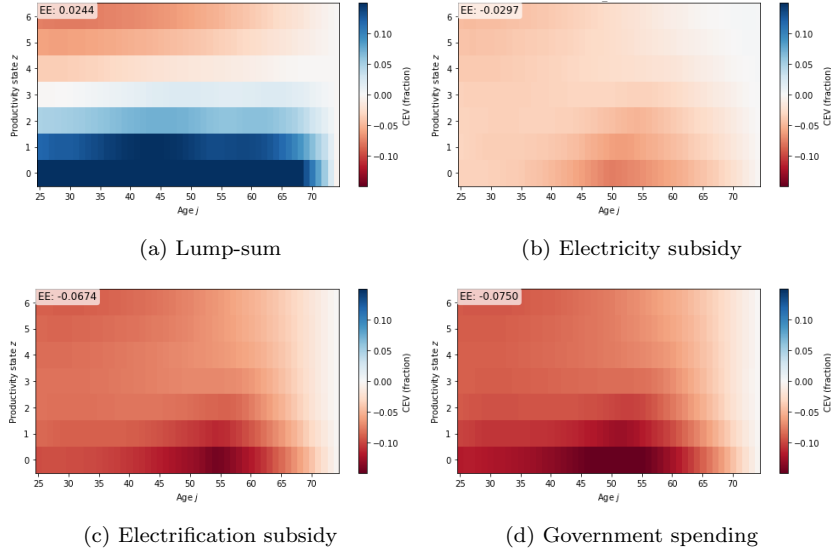


Figure 19: Replication of figure 6 without income risk

EoS energy and non-energy factors, ν A higher elasticity of substitution would imply more substitutability between energy and the capital-labor input in the production of the final consumption good. In this case, an increase in energy prices is expected to have a less severe negative effect on GDP and wages. I reanalyze GDP and wage decreases when adjusting ν from 0.2 to 0.4 but while keeping the carbon tax constant. This allows for a clean comparison of the robustness of the economic distortions caused by the tax with respect to the EoS between energy and other inputs. Figure 20 shows that results stay the same quantitatively and qualitatively.

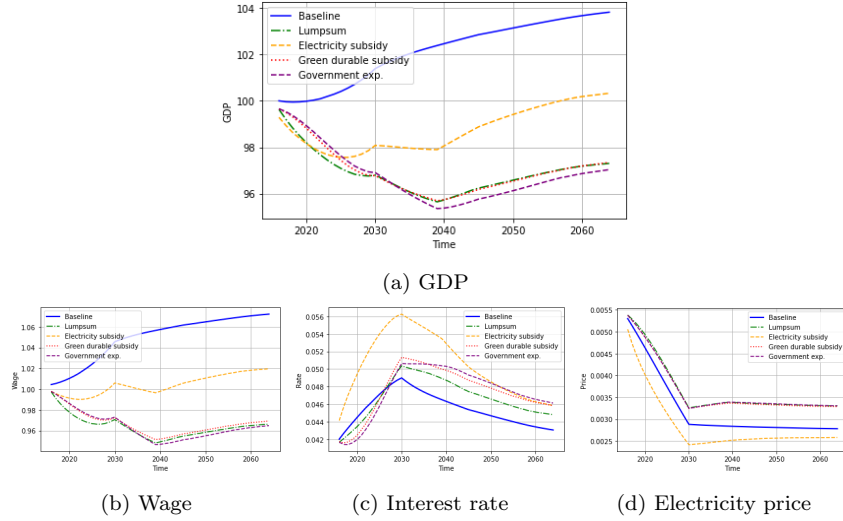


Figure 20: Replication of figure 5 with elasticity of substitution between energy and other inputs, ν , equal to 0.4.

EoS renewable and fossil electricity, λ The elasticity of substitution governs the extent to which the electricity firm can substitute towards renewable sources when the carbon tax rises. A higher λ is therefore expected to lead to a lower peak tax rate, but the degree to which it does could depend on the instrument, i.e. whether or not renewable electricity is subsidized. In order to answer this question, I analyse the cases of a high and low λ , namely 5 and 0.5, in line with ranges used in Hochmuth et al. (2025). In figure 21, I compare tax rates in the baseline regime ($\lambda = 1.8$), with a high and low elasticity regime.

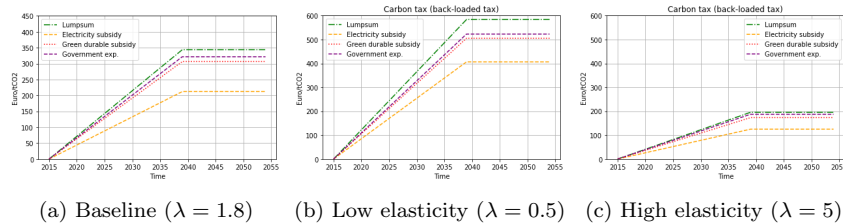


Figure 21: Tax rate under several EoS

Results are in line with findings from Hochmuth et al. (2025), who find that this elasticity of substitution is an important determinant of the endogenous tax rate required to transition to a low-emission economy. Importantly, I do not find any interaction effects with the instruments. That is, the effect of λ on the maximum tax rate is the same for each instrument. This ensures the robustness of assessing between instruments.

Down-payment constraint, λ^{LTV} The salience of the financing frictions is determined by the down payment constraint. In the current calibration, a 10% down-payment is required. I check robustness of the results with alleviating the down-payment constraint, that is if everyone is allowed to borrow against the full amount against the market interest rate, implying a risk premium of 0%.

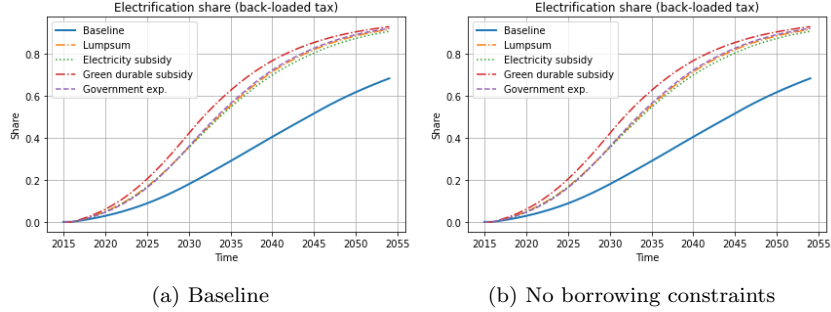


Figure 22: Household electrification

This shows that few households are affected by alleviating constraints. This can be explained by the fact that borrowing is still expensive, as interest rates rise sharply over the transition (figure 5, panel (c)). The high cost of borrowing makes that alleviating risk premia or increasing the space of borrowing does not affect borrowing at the margin. It does make borrowing for households who are already taking out credit cheaper. This means that the effects for the lower-end of the income distribution who do not electrify are not driven by arbitrary assumptions on the down-payment constraint.